



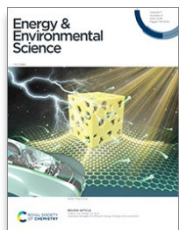
Spatially explicit assessment of CCS and hydrogen infrastructure in China enables 7.5 GtCO₂ storage by 2050

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Complete List of Authors:	Acen, Caroline ; Chengdu University of Technology, Sichuan Industrial Intelligent Monitoring and Application Engineering Technology Research Centre Bamisile, Olusola ; Chengdu University of Technology, Sichuan Industrial Intelligent Monitoring and Application Engineering Technology Research Centre; Imperial College London, Centre for Environmental Policy; University of Dundee, Energy and Environmental Science Division Olasehinde, Daniel ; Chengdu University of Technology, Sichuan Industrial Intelligent Monitoring and Application Engineering Technology Research Centre Rogelj, Joeri; Imperial College London Olagundoye, Omosalewa ; Chengdu University of Technology, Sichuan Industrial Intelligent Monitoring and Application Engineering Technology Research Centre Ejiyi, Chukwuebuka ; Chengdu University of Technology, Sichuan Industrial Intelligent Monitoring and Application Engineering Technology Research Centre Staffell, Iain; Imperial College London, Centre for Environmental Policy Huang, Qi; Chengdu University of Technology, Sichuan Industrial Intelligent Monitoring and Application Engineering Technology Research Centre

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Dear Professor Wilcox, the Editorial Board,

We are pleased to enclose “*Spatially explicit assessment of CCS and hydrogen infrastructure in China enables 7.5 GtCO₂ storage by 2050*” for your consideration.

Carbon capture and storage is now central to many national and global net-zero strategies, but its practical deployment depends on where storage is located relative to emissions sources, how quickly CO₂ can be injected, and whether geological risks can be managed. These constraints are especially important for China, the world’s largest emitter, where CCS may play a major role in decarbonizing hard-to-abate industries and supporting low-carbon hydrogen production.

We develop a spatially explicit national assessment of China’s CCS and low-carbon hydrogen infrastructure potential. We integrate high-resolution geological storage data, injection-rate constraints, full-chain CCS cost modelling, source-sink matching, a comparison of transport modes, and seismic risk assessment. We assess the trade in emissions sequestration between provinces and the trade-offs between different types of storage site. This allows us to move beyond aggregate estimates of theoretical storage capacity and identify where CCS can be deployed safely, economically and at sufficient scale.

The manuscript makes three main advances. First, it quantifies the practical distinction between “big-but-slow” deep saline aquifers and “small-but-fast” enhanced oil recovery resources. Second, it identifies priority CCS hubs and cross-provincial transport requirements by linking industrial emissions to viable storage sites via an optimized network. Third, it extends CCS infrastructure planning to blue hydrogen, identifying regions where natural gas resources, industrial demand and CO₂ storage can be co-located to reduce infrastructure costs. Our analysis shows that China could store 7.5 GtCO₂ in deep saline aquifers by 2050, but also finds that injectivity, seismic risk and provincial mismatches create binding constraints that are obscured in national-scale capacity estimates.

We believe the paper is well suited to Energy & Environmental Science because it addresses a major climate and infrastructure challenge through an interdisciplinary framework. As such, it will be of interest beyond CCS specialists, also to readers concerned with industrial decarbonization, hydrogen strategy, climate mitigation pathways, infrastructure planning and the governance of net-zero transitions.

We confirm that this manuscript is original, has not been published previously, is not under consideration elsewhere, and has been approved by all authors.

Thank you for your consideration.

Yours sincerely,
Iain Staffell
on behalf of all authors

Broader context

Reaching net zero will require more than replacing fossil-fuel power stations with renewable electricity. Many essential industries, from steel and cement to chemicals and hydrogen production, will still release carbon dioxide in ways that are difficult or costly to eliminate. Carbon capture and storage could help manage these residual emissions, but only if captured CO₂ can be moved to suitable underground reservoirs safely, affordably and at sufficient scale. This study shows why national estimates of storage potential are not enough by themselves. The location of emitters, pipelines, injection rates, geological risks and future demand must also align. Focusing on China, the world's largest CO₂ emitter, we find that large-scale storage is possible but unevenly distributed, creating clear winners, bottlenecks and coordination challenges. Deep saline aquifers could store 7.5 GtCO₂ by 2050, while enhanced oil recovery provides a smaller near-term option. Pipeline networks are consistently cheaper than tanker-based transport, and many industrial sources lie close enough to viable storage sites to support regional hubs. These findings provide an academic basis for turning carbon-management ambitions into a spatially-explicit infrastructure roadmap, helping policymakers and industry decide where to invest first, where risks require the closest management, and how CCS can support a credible transition for hard-to-abate sectors.

204 words.

Spatially explicit assessment of CCS and hydrogen infrastructure in China enables 7.5 GtCO₂ storage by 2050

**Caroline Acen¹, Olusola Bamisile^{*1,2,3}, Daniel Olasehinde¹, Joeri Rogelj⁴,
Omosalewa Olagundoye¹, Chukwuebuka Ejiyi¹, Qi Huang¹, Iain Staffell^{*2}**

¹ Sichuan Industrial Intelligent Monitoring and Application Engineering Technology Research Centre, College of Nuclear Science and Automation Engineering, Chengdu University of Technology, Sichuan, 610059, P.R. China.

² Centre for Environmental Policy, Imperial College London, United Kingdom.

³ Energy and Environmental Science Division, CEPMLP, University of Dundee, Scotland, United Kingdom.

⁴ Grantham Institute for Climate Change and Environment, Imperial College London, United Kingdom.

* Corresponding author: obamisile@dundee.ac.uk, i.staffell@imperial.ac.uk

Abstract

Carbon capture and storage (CCS) is central to climate change mitigation for sectors that cannot easily electrify. China, the world’s largest emitter, has ample geological CO₂ storage and low-carbon hydrogen potential, yet national estimates rarely identify sites that are safe, low-cost, and fast enough to meet climate targets. Here, we present a national, 5-km resolution geospatial assessment of Chinese CCS deployment potential, accounting for injection constraints and seismic risk in deep saline aquifers and enhanced oil recovery. We quantify rate-constrained CO₂ removal costs and optimize spatial matching of industrial sources, blue hydrogen, transport, and sinks. Storage potential is large but uneven, requiring inter-provincial transport. China’s technical sequestration potential reaches 1.26 GtCO₂ by 2030 and 7.5 GtCO₂ by 2050 in aquifers, while oil recovery is smaller but faster: 1.08 and 0.82 GtCO₂, respectively. The lowest full-chain CCS cost of 838 USD/km in 2025 declines to 347 USD/km by 2050 via economies of scale and technological progress. Pipeline transport is 14–29% cheaper than tankers, and 89% of manufacturing sources lie within 250 km of viable storage. Contrary to aggregate estimates finding ample capacity, injectivity, seismic risk and provincial mismatches impose binding constraints. These results identify priority carbon-storage and blue-hydrogen hubs, offering a spatial roadmap for China’s 2060 carbon-neutrality pledge.

Introduction

Limiting global warming to below 1.5°C without overshoot is increasingly unlikely¹. Scenarios consistent with the Paris Agreement increasingly rely on carbon dioxide removal (CDR) to offset residual emissions and reduce temperature overshoot². Geological carbon storage can provide the long-term sequestration required to verify permanent removal³. This intersects with hard-to-abate industries, including aviation, shipping, trucking, steel, cement, aluminium, and chemicals, which contributed ~40% of global greenhouse gas (GHG) emissions in 2024, and lag behind the power and light transport sectors in decarbonization^{4,5}. For these industries, carbon capture, utilization, and storage (CCUS) offers a dual function: abating process emissions at source and enabling permanent storage of biogenic or atmospheric CO₂.

Current climate mitigation scenarios consistently assign a substantial role to CCUS. Most pathways to global net-zero by 2050 feature large CCUS deployment, rising from ~1-5 GtCO₂ per year in 2030 to ~5-13 GtCO₂ by 2050^{6,7}. IPCC scenarios present a wide range of CCUS deployment, reducing emissions by 0-5 GtCO₂ by 2030 (median: 1 GtCO₂), expanding to 0-13 GtCO₂ by 2050 (median estimate: 2-3 GtCO₂)⁷. However, these scenarios predominantly assume mitigation actions beginning in 2020, with immediate global scale-up. As this has already been missed, achieving even 2030's median deployment is now less feasible, and later scale-up scenarios warrant greater attention. Current CCUS deployment is nascent, with ~50 facilities operating globally in 2025 capturing 0.063 GtCO₂^{10,11}. A further 44 projects are under construction and 480-534 in planning^{12,13}. If all proceed, global capacity could exceed 0.5 GtCO₂ by 2030^{14,15}, representing 1.1% of global total CO₂ emissions in 2024^{1,2}.

Geologic carbon storage has substantial technical potential, with estimates ranging from 10,000 to 40,000 GtCO₂^{16,17}. A risk-based planetary assessment suggests a prudent global storage limit of ~1,460 GtCO₂¹⁸. National analysis for China indicates that onshore basins alone could store 448 GtCO₂, enough to sequester national cumulative emissions from 1990 to 2060¹⁹. China is the world's largest GHG emitter, producing 29% of global CO₂ emissions²⁰. It pledged to peak emissions by 2030 and achieve carbon neutrality by 2060^{21,22}. As a leading producer and consumer of coal (4.76 and 4.9 Gt in 2024)^{24,23}, CCUS will be critical for decarbonizing China's hard-to-abate sectors while balancing energy security and economic growth²⁵. Currently, China hosts 36 CCUS projects^{12,13}, of which 64% are operational, 22% under construction, and 14% in planning¹³. Current capture capacity remains limited to ~0.001 GtCO₂ per year^{12,13}. Coal-based projects are prioritized, with post-2030 capture projected to reach 0.040-0.043 GtCO₂ and storage capacity reaching 0.003 GtCO₂/year^{12,13}. Initially driven by industrial CO₂ utilization in food processing and manufacturing, China's CCUS development is now motivated by climate objectives.

To align with the 2060 carbon neutrality pathway, China must expand CCUS capacity to 0.2 GtCO₂/year by 2030 and 2.7 GtCO₂/year by 2050²⁵. Nonetheless, large-scale carbon storage deployment carries serious risks, given uncertainty about storage capability and injection rates^{26,27}.

While research has improved CO₂ storage capacity estimates, five limitations persist which reduce their practical utility and risk misinforming investment and policy decisions. First, CCS planning often omits geological storage potential coupled with injection rate constraints²⁸. This overlooks the link between storage volume and the rate at which CO₂ can be safely sequestered: a primary determinant of site value and cost. Second, risk assessment for geological containment is insufficiently considered. Potential CO₂ leakage through undetected or reactivated faults, and induced seismicity from prolonged fluid injection^{29,30}, pose environmental and social challenges. These are frequently addressed only qualitatively, if at all, rather than through integrated, quantitative geospatial analysis^{31,32}. Third, analyses typically neglect long-term effects of extreme events, socio-economic change, and spatial heterogeneity in source-sink distribution^{33,34}. Large distances between major emitters and optimal storage basins create economic and logistical barriers that national-scale aggregate analyses gloss over (see Appendix 1). Fourth, many studies focus on few hard-to-abate sectors (power, iron and steel, chemicals, cement), neglecting other industries needed to meet climate goals^{35,36}. Finally, most studies do not fully consider secondary CO₂ transport within individual emission hubs or the potential evolution of CCUS infrastructure layout over time³⁷. These gaps highlight the need for comprehensive CCS planning frameworks that integrate geological capacity, injectivity dynamics, geomechanical risks, and socio-technical constraints into a unified model for strategic investment and policymaking.

Here we present a modelling framework that combines techno-economic modelling, geospatial assessment, source-sink matching, and proximity analysis. We use China's high-resolution grid-level datasets for several applications: quantifying investment costs for CO₂ sequestration at potential storage sites through spatial mapping; establishing optimal coupling between emission sources (industrial clusters) and storage sinks using fine-grid resolution analysis; assessing provincial-level contributions of CCS to emission reduction; identifying suitable locations for blue hydrogen facilities co-located with CO₂ storage reservoirs; and evaluating geological storage risks by analysing active fault lines to determine areas with higher potential for CO₂ leakage or induced seismicity. Although China possesses substantial CO₂ storage capacity, deployment faces challenges due to the heterogeneous spatial distribution of storage resources, injectivity limitations, and considerable distances between suitable storage sites and major emission sources (see Appendix 2). Our work provides a high-resolution (5×5 km) spatial and economic evaluation of China's CCS potential, linking geological assessments with practical decarbonization pathways.

Methods

This study uses a high-resolution dataset of carbon geological storage potential and injection rate capacity for deep saline aquifers (DSA) and enhanced oil recovery (EOR) systems located onshore and offshore across China. With a grid resolution of 5 km x 5 km, the dataset encompasses 1,181 oilfields and 69 reservoirs distributed within 24 major sedimentary basins²⁸. A comprehensive assessment of

China's CO₂ storage resources is made possible by this dataset's high spatial resolution, which can be scaled from localized grid-level analysis to more extensive assessments at the county, provincial, and basin-wide levels²⁸.

Economic analysis

We model the full-chain costs of CCS, encompassing capture, transportation (via truck or pipeline), and storage. Cost estimates for each stage, derived from an analysis of existing CCUS projects in China, are presented in Table 1⁷¹ (see Supplementary Table 1 for the original costs in Chinese Yuan (CNY)).

Table 1: The economic cost of each component of the full-chain CCS system^{59,70,71}. 1 USD = 7.183 CNY.

Year		2025	2030	2035	2040	2045	2050
Capture Cost (USD/tonne)	Pre-combustion	13.92 - 25.06	12.53 - 18.10	9.75 - 11.14	6.96 - 9.75	4.18 - 6.96	2.78 - 5.57
	Post-combustion	32.02 - 43.16	26.45 - 38.98	22.27 - 30.63	13.92 - 25.06	11.14 - 20.88	9.75 - 16.71
	Oxy-fuel combustion	41.77 - 66.82	22.27 - 54.29	18.10 - 44.55	15.31 - 32.02	12.53 - 20.88	11.14 - 18.10
Transportation (USD/tonne.km)	Tank	0.13 - 0.19	0.11 - 0.18	0.10 - 0.17	0.08 - 0.15	0.07 - 0.15	0.07 - 0.14
	Pipeline	0.11	0.10	0.08	0.07	0.07	0.06
Storage cost (USD/tonne)		6.96 - 8.35	5.57 - 6.96	4.87 - 5.57	4.18 - 4.87	3.48 - 4.18	2.78 - 3.48

Given the prevailing adoption of post-combustion capture technology in CCUS deployments across the globe, we base our cost calculations on this method to ensure consistency with real-world applications⁷⁰. Furthermore, we evaluate three cost scenarios reflecting the minimum, maximum, and average unit costs, derived from the ranges presented in Table 1, as shown in Table 2 (see Supplementary Table 2 for the original costs in CNY).

Table 2: Minimum, maximum, and average costs of each segment of the full-chain CCS system.

Year	Post-Combustion Capture (USD/tonne)	Tank Transport (USD/tonne.km)	Pipeline Transport (USD/tonne.km)	Storage (USD/tonne)
	Min / Mean / Max	Min / Mean / Max	Fixed	Min / Mean / Max
2025	32.02 / 37.59 / 43.16	0.13 / 0.16 / 0.19	0.11	6.96 / 7.66 / 8.35
2030	26.45 / 32.72 / 38.98	0.11 / 0.15 / 0.18	0.10	5.57 / 6.27 / 6.96
2035	22.27 / 26.45 / 30.63	0.10 / 0.13 / 0.17	0.08	4.87 / 5.22 / 5.57
2040	13.92 / 15.31 / 25.06	0.08 / 0.12 / 0.15	0.07	4.18 / 4.53 / 4.87
2045	11.14 / 16.01 / 20.88	0.07 / 0.11 / 0.15	0.07	3.48 / 3.83 / 4.18
2050	9.75 / 13.23 / 16.71	0.07 / 0.10 / 0.14	0.06	2.78 / 3.13 / 3.48

The economic model evaluates the total investment cost for the full-chain CCS system in each grid cell i according to Equation 1:

$$Cost_{t,i} = uc_t \times CCS_{cap,t,i} \quad (1)$$

where uc_t represents the unit abatement cost in year t , comprising the sum of capture (C), transportation (T), and storage (S) unit costs, and $CCS_{cap,t,i}$ denotes the annual volume of CO₂ captured and stored in grid cell i . For modelling purposes, we equate $CCS_{cap,t,i}$ to the injection rate (R_i) at each grid cell.

Based on established literature, the distance (d) between emission sources and storage sites must be within 250 km, beyond which an additional compressor station to maintain supercritical flow is required (see [Supplementary Table 3](#))⁷⁰. Analysis indicates that integrating such infrastructure increases transportation costs by 30% to 100%, contingent upon pipeline-specific design parameters, total distance, and terrain complexity^{72,73}. To model the economically optimal scenario, the full-chain cost per kilometre is therefore calculated for distances (d) not exceeding this 250 km threshold, as defined by Equation 2:

$$Cost_{t,i} = \left(\frac{(C + S) \times R_i}{d} \right) + T \times R_i \quad (2)$$

Where R_i is the injection rate. To ensure unit consistency, the injection rate (Mt/year) and storage potential (Mt) are converted to tonnes. The geological storage capacity represents a finite resource within each grid cell. Consequently, under the assumption of zero leakage (Equation 3), the remaining storage potential diminishes annually (see [Supplementary Table 4](#)):

$$S_{i,t} = S_{2025} - R_i \times (t - 2025) \quad (3)$$

where S_{2025} indicates the initial storage capacity. Subsequent annual injection rates beyond 2025 are determined through Equation 4:

$$R_{i,t} = \min(R_i, S_{i,t}) \quad (4)$$

Seismic risk factor calculation

In alignment with recent methodological frameworks for assessing geological storage resources under varying degrees of risk aversion, such as that proposed by Gidden et al. (2025)¹⁸, a differentiated potential comprising precautionary, advanced, and full techno-economic estimates was considered. However, implementing such a tiered classification was not feasible for this analysis, as the underlying storage potential dataset (from ²⁸) was published with exclusion layers, including

population centres, protected areas, and economic zones, already integrated. Accordingly, the dataset represents a consolidated maximum storage potential within predefined basin and oilfield boundaries, precluding the retrospective disaggregation of individual exclusion criteria to construct risk-tiered estimates. The seismic risk factor for EOR and DSA carbon storage sites across China is derived through a spatially explicit, probabilistic multi-hazard framework that integrates seismic ground motion hazard, tectonic proximity, and site-specific geotechnical vulnerability. This approach synthesizes the primary physical mechanisms contributing to seismic risk for subsurface storage infrastructure: the intensity of shaking expected at the site, the influence of nearby active faults, and the potential amplification of ground motions due to local soil conditions⁷⁴. The final seismic risk factor, R , is formulated as a weighted linear combination of three normalized component scores as expressed in Equation 5:

$$R = w_1 \times S_{PGA} + w_2 \times S_{FP} + w_3 \times S_{SA} \quad (5)$$

where S_{PGA} is the peak ground acceleration hazard score, S_{FP} is the fault proximity risk score, and S_{SA} is the site amplification vulnerability score. The weighting coefficients w_1 , w_2 , w_3 satisfy $w_1 + w_2 + w_3 = 1$, and reflect the relative importance assigned to each risk driver. In this national-scale assessment, a weighting scheme of (0.5, 0.3, 0.2) is adopted, following the implicit prioritization found in seismic design codes and hazard assessment guidelines⁷⁵, where PGA serves as the primary ground motion parameter^{76,77}, while fault proximity and site conditions are treated as modifying factors^{78,79}. This additive model provides a transparent and computationally efficient metric for comparative risk ranking across hundreds of fixed-location storage sites. Acknowledging that more complex multiplicative or nonlinear interactions may exist, the linear form offers stability and clarity for a first-order national assessment.

The ground motion hazard score, S_{PGA} , is calculated using probabilistic seismic hazard analysis (PSHA) outputs for mainland China. We utilize the time-dependent PSHA developed by Xu et al.⁸⁰, which integrates multiple seismic source models, including area sources, fault sources, and seismic risk areas, with updated seismicity parameters for the period 2021–2030. PGA values, representing the severity of expected shaking, are extracted directly at the coordinates of each of the 527 storage sites from hazard maps for six exceedance probabilities (63%, 10%, 5%, 2%, 1%, and 0.5% in 50 years) (Supplementary Figure 1). To render these absolute acceleration values into a dimensionless, comparable risk score, a spatial normalization is applied (Equation 6):

$$S_{PGA}(p) = \left(\frac{PGA(p)}{PGA_{max}(p)} \right) \times 10 \quad (6)$$

Where $PGA(p)$ is the site-specific PGA value at probability level, p , and $PGA_{max}(p)$ is the maximum PGA value observed across all of China for that same probability. This transformation scales the score from 0 to 10, preserving the relative spatial distribution of seismic hazard across the country. The 10%

in 50-year probability (475-year return period) hazard level serves as the primary basis for the risk calculations, representing a standard for important infrastructure design⁸⁰.

The fault proximity risk score, S_{FP} , quantifies the heightened risk associated with a storage site's vicinity to active tectonic structures. This component integrates both the intrinsic hazard potential of the nearest fault and the geometric decay of influence with distance. The intrinsic fault hazard is characterized by a Fault Activity Risk Score (FARS) (see Supplementary Figure 2), calculated for each fault segment in the China Active Fault Database (CAFD400) based on its most recent activity and kinematic behaviour (Table 3)⁸¹. The FARS is computed as:

$$FARS = (W_{age} \times 1.5) + W_{fea} \tag{7}$$

where W_{age} is a weight assigned based on the fault's latest active period (see Table 3)⁸², and W_{fea} is a weight based on fault type and detectability (with buried or inferred faults receiving the highest weight due to greater epistemic uncertainty). The multiplier of 1.5 amplifies the influence of recent activity, reflecting its critical role in contemporary hazard.

Table 3: Age-weight assignment based on the latest active period and fault property weight (W_{fea}) based on kinematic type and detectability.

Age		Fault Property	
Latest Active Period	Age Weight (W_{age})	Fault Type Category	Fault Property Weight (W_{fea})
Holocene (Qh)	4	Buried / Inferred (any subtype)	4
Late Pleistocene (Qp3)	3	Complex oblique-slip (e.g., left/right lateral normal, normal-strike-slip, reverse strike-slip)	3
Middle-Early Pleistocene (Qp1+2)	2	Simple slip (normal, reverse, strike-slip, left/right lateral)	2
Pre-Quaternary (pre-Q)	1	Low-hazard types (exposed, flexural slip, tear fault)	1

For a storage site located a distance, d , (in kilometres) from a fault with a given FARS, the fault proximity score is modelled using an exponential decay function as depicted in Equation 8:

$$S_{FP}(p) = FARS \times \exp\left(-\frac{d}{\lambda}\right) \tag{8}$$

The decay constant, λ , defines the characteristic distance over which the fault influence diminishes. A default value of $\lambda=200$ km is used, consistent with the typical zone of influence for major continental faults. This formulation captures the intuitive principle that risk is highest when sitting directly on a recently active fault, and seismic wave amplitude decreases smoothly with distance due to geometric spreading and anelastic attenuation^{83,84}.

The site amplification vulnerability score, S_{SA} , accounts for the modification of bedrock ground motions by local near-surface geology. The key parameter for estimating this amplification is the time-averaged shear-wave velocity in the upper 30 metres, V_{S30} . For this study, we employ a newly developed, high-resolution V_{S30} map for mainland China at 30-arcsecond resolution, generated using a refined Slope-based Cokriging (SCK) model (Supplementary Figure 3)⁸⁵. This model spatially integrates 7,939 local V_{S30} measurements (7,797 boreholes plus 142 strong-motion stations) with topographic slope data via a multivariate geostatistical framework. The SCK estimator at a target location, x_0 , is given by Equation (9):

$$V_{S30}^{SCK}(x_0) = \sum_{\alpha \in S_{VS}} \lambda_{vs}(\alpha) V_{S30}(\alpha) + \sum_{\beta \in S_{Slp}} \lambda_{slp}(\beta) Slp(\beta) \quad (9)$$

where λ_{vs} and λ_{slp} are weights solved through a Cokriging system that models the spatial autocorrelation and cross-correlation between V_{S30} and slope. This method provides a more accurate and spatially continuous estimate than global slope proxies, particularly in complex terrains. The derived V_{S30} value at each storage site is then used in the empirical ground motion model by Boore et al.⁴¹ to calculate a linear site amplification factor, F , as expressed in Equation 10:

$$\ln(F) = b \times \ln\left(\frac{V_{S30}}{V_{ref}}\right) \quad (10)$$

where $b = -1.186$ and $V_{ref} = 760$ m/s. The amplification factor is subsequently transformed into a normalized vulnerability score bounded between 0 and 10:

$$S_{SA} = \left(\frac{F - 0.5}{3.0 - 0.5}\right) \times 10 \quad (11)$$

where the bounds of 0.5 and 3.0 for F are based on empirical limits observed in global ground motion databases⁴¹. This scoring reflects higher vulnerability (higher score) for softer sites with lower V_{S30} that exhibit greater amplification.

To rigorously evaluate the robustness of the resulting risk rankings and to quantify the influence of epistemic uncertainties in model parameters, a comprehensive sensitivity analysis is conducted. This analysis systematically varies three key inputs: the fault influence decay distance λ (across 50, 100, 200, 300, and 500 km), the PGA hazard exceedance probability (across all six available levels), and the site amplification methodology (comparing the Boore et al. model⁴¹, the Chinese seismic code GB 50011-2010 approach⁴², and a no-amplification baseline). For each sensitivity scenario, the complete risk calculation (Equation 1) is recomputed, producing 14 distinct risk assessments (5 fault λ values + 6 PGA levels + 3 amplification methods) in addition to the base case. This approach enables

quantification of parameter influence through comparison of resulting risk scores and spatial risk patterns.

A key limitation of this study is its inability to categorize storage potential by risk tolerance levels due to the aggregated nature of the underlying datasets used. While the seismic risk analysis provides an important contribution, the storage dataset's integrated exclusion layers prevent a stepwise, risk-tiered assessment.

Spatial coupling of emission sources and storage sinks

To identify regions with the highest potential for synergistic alignment between CO₂ emission sources (industrial clusters) and geological storage sinks across China, we use a gridded manufacturing dataset (GMD)⁸⁶ that maps Chinese industrial facilities at 1×1 km resolution. This dataset lists manufacturing firms across eight distinct industrial categories for both 2015 and 2019 (Supplementary Table 5). Our analysis specifically uses the 2019 data, processed according to the same methodological framework described earlier. Through geospatial overlay analysis, we systematically match emission sources with potential storage sites, focusing on identifying optimal pairings where sources and sinks are within a 250 km radius (Supplementary Figure 4). This proximity constraint considers practical aspects of cost-effective CO₂ transport infrastructure development.

Spatial coupling of blue hydrogen production and storage sinks

Our analysis identifies optimal locations for blue hydrogen production by integrating CO₂ storage potential with existing energy infrastructure and industrial capacity. We extract China-specific oil and gas field data from the Global Oil & Gas Features Database⁸⁷, applying consistent processing methods as previously described. From the industrial manufacturing classification scheme (GMD), we focus on hydrogen-relevant manufacturing sectors, including: Mechatronics and Equipment (ME19), Metallurgical Chemical Industry and Resource Processing (MC19), and Pharmaceutical Manufacturing (PM19)⁸⁶. Through advanced GIS techniques, we conduct multi-criteria spatial analysis incorporating: geological storage capacity, natural gas field distribution, and hydrogen-related manufacturing industrial clusters. The methodology employs proximity-based ranking, identifying candidate locations where all three critical elements fall within a 250 km radius, a distance constraint reflecting practical economic considerations for hydrogen production and carbon management infrastructure.

The production of blue hydrogen via conventional steam methane reforming (SMR) is characterized by a significant carbon footprint, generating approximately 10–11 tonnes of CO₂ for every tonne of hydrogen produced⁸⁸. This results in a CO₂-to-H₂ mass ratio of roughly 10:1. The potential annual blue hydrogen production capacity for China by mid-century was estimated by linking the national CO₂ geological storage potential with the stoichiometric and energetic requirements of steam

methane reforming (SMR) with CCS. The calculation is based on the premise that the scale of blue hydrogen production is directly constrained by the available and allocated CO₂ storage capacity. The total storage resource allocated for hydrogen, S_{H_2} (in MtCO₂), was derived from a fraction of the national technical potentials for EOR and DSA, as defined in our study (Equation 12):

$$S_{H_2} = \alpha \times S_{EOR} + \beta \times S_{DSA} \quad (12)$$

where S_{EOR} and S_{DSA} are the total technical sequestration potentials (0.8226 and 7.5 GtCO₂ by 2050, respectively), and α (0.3) and β (0.05) are the allocation factors representing the portion of each resource dedicated to hydrogen production. The allocation factors for carbon capture capacity are based on current deployment data and announced projects. As of 2025, China hosts 23^{12,13} of the 50 operational CCUS projects globally (46%)^{10,11}. Globally, 30% of announced CCUS capacity is linked to hydrogen production⁵², equivalent to approximately 150 MtCO₂/year out of a global total of 500 MtCO₂/year by 2030^{14,15}. In China, blue hydrogen projects comprise five operational facilities (1.4 MtCO₂/year) and one proposed project (1.0 MtCO₂/year), resulting in a potential capture capacity of 1.4–2.4 MtCO₂/year by 2030. This represents only 0.9–1.6% of global hydrogen-linked CCUS capacity, indicating the early stage of blue hydrogen development in China despite its high overall project count. Using a CO₂-to-H₂ ratio of 10:1, this translates to blue hydrogen production of 0.14–0.24 Mt H₂/year by 2030, with projections to 2050 based on announced pipelines. To avoid overreliance on any single storage resource⁸⁹, the study adopts allocation fractions of 10% ($\alpha = 0.10$) of EOR technical potential⁹⁰ and 5% ($\beta = 0.05$) of DSA technical potential^{91,92}. These conservative estimates are justified by China's modest global share of hydrogen-linked CCUS (0.9–1.6%), the need to reserve storage capacity for other hard-to-abate sectors^{90,93,94}, and the higher uncertainty and costs associated with DSA deployment^{95,96}.

To test the influence of these parameters on production capacity and hub deployment, a sensitivity analysis is presented in Table 4. Under low-allocation, slow-deployment assumptions ($\alpha = 0.05$, $\beta = 0.02$, $k = 1.2$), production capacity declines to 0.8 Mt H₂/year, corresponding to approximately 4 to 8 hubs. Under high-allocation, accelerated-deployment assumptions ($\alpha = 0.20$, $\beta = 0.05$, $k = 2.5$), capacity reaches 4.5 Mt H₂/year, corresponding to approximately 22 to 45 hubs. If DSA is also deployed at higher rates ($\alpha = 0.20$, $\beta = 0.10$, $k = 2.5$), capacity could rise to 7.6 Mt H₂/year under the most optimistic scenario, corresponding to approximately 38 to 76 hubs. The central range of 2.5–3.0 Mt H₂/year ($\alpha = 0.10$, $\beta = 0.05$, $k = 1.5$ –2.0) and 15–25 hubs is presented as the main finding, with the recognition that actual deployment will depend on competition for storage capacity among industrial sectors, the pace of deep saline aquifer characterization and permitting, and the realization of announced project pipelines.

Table 4. Sensitivity analysis parameters and resulting blue hydrogen production capacity by 2050. Assuming 0.10–0.20 Mt H₂/year per major hub, consistent with large-scale SMR-CCS facility benchmarks^{57,58}.

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α	β	k	S_{H_2} (MtCO ₂)	H_{avg} (Mt H ₂ /year)	H_{2050} (Mt H ₂ /year)	Estimated hubs*
0.05	0.02	1.2	191	0.64	0.8	4–8
0.10	0.05	1.5	457	1.52	2.3	12–23
0.10	0.05	2.0	457	1.52	3.0	15–30
0.15	0.05	2.2	498	1.66	3.7	18–37
0.20	0.05	2.5	539	1.80	4.5	22–45
0.20	0.10	2.5	914	3.05	7.6	38–76

This total allocated storage was then converted into an average annual hydrogen production capacity, H_{avg} (MtH₂/year), over an assumed project lifespan, L , of 30 years, using the standard CO₂-to-H₂ mass ratio, $r = (10:1)$, for SMR-CCS (Equation 13):

$$H_{avg} = \frac{S_{H_2}}{L \times r} \tag{13}$$

To reflect the projected infrastructure ramp-up and operational maturity by 2050, the average annual production was scaled by a dynamic deployment factor, k (1.5–2.0), yielding our final estimated capacity range for mid-century (Equation 14):

$$H_{2050} = k \times H_{avg} \tag{14}$$

The deployment factor k is derived empirically from observed and projected growth rates for fossil-based hydrogen with CCUS. Global fossil fuel with CCUS hydrogen production grew from 0.57 Mtpa in 2021 to 0.73 Mtpa in 2025⁵². Projections indicate a sixfold increase in total low-emission hydrogen from 0.8 Mtpa in 2024 to 4.2 Mtpa by 2030⁵², with blue hydrogen maintaining approximately 70% share⁵², implying growth from 0.67 Mtpa to ~2.9 Mtpa over the same period. Extrapolating this S-curve trajectory to 2050, where total Chinese hydrogen demand is expected to reach 60 Mtpa^{97–99}, blue hydrogen production of 2.5–4.0 Mtpa corresponds to 4–7% of total demand. This trajectory yields a ratio of mid-century capacity to the 30-year linear average ($H_{avg} = 1.52$ Mtpa) of $k = 1.6$ –2.6. We therefore adopt $k = 1.5$ –2.0 as the central range. Sensitivity analysis varies k from 1.2 (slow deployment) to 2.5 (accelerated deployment), α from 0.05 to 0.20, and β from 0.02 to 0.10.

Provincial contribution to emission reduction

Our analysis evaluates provincial-level contributions to carbon emission reductions across China, as quantified by Equation 15:

$$C_{j,t} = \frac{(t - 2025) \times R_{j,t}}{emis_{j,t}} \times 100 \tag{15}$$

where $C_{j,t}$ represents the CCS-mediated emission reduction for province j , $R_{j,t}$ denotes the province-specific average injection rate, and $emis_{j,t}$ corresponds to provincial carbon emissions in year t . We employ injection rates rather than CCS facility capture capacities in our calculations, reflecting the operational reality that most facilities transport captured CO₂ across provincial boundaries for geological storage¹³. The model incorporates storage depletion dynamics while maintaining a fixed upper bound for cumulative sequestration capacity equivalent to the maximum geological storage potential. Provincial carbon emission projections from 2025 through 2050 were sourced from Ref. ⁴⁹, while **Supplementary Table 6** presents the aggregated data on provincial storage capacities and corresponding average injection rates²⁸.

Results and discussion

Estimating the carbon removal cost curve

We calculate the full-chain CCS costs across 12 scenarios with minimum, average, and maximum unit cost estimates for capture, transport, and storage/injection phases. These scenarios consider both pipeline and tanker transportation modes for enhanced oil recovery (EOR) and deep saline aquifer (DSA) storage options (see **Supplementary Table 7–10**). Our analysis reveals that pipeline transport is more cost-effective for CCS than tanker-based systems within a 250 km radius of storage sinks (**Supplementary Figures 5–16**). **Figure 1** and **Figure 2** illustrate the spatial distribution of average full-chain CCS costs at 5×5 km resolution across all scenarios, demonstrating how geological storage potential and injectivity constraints influence CCS economics throughout China from 2025 to 2050.

The full-chain CCS cost across China exhibits spatial and temporal heterogeneity. In 2025 (**Figure 1a**), costs are notably higher in several regions, especially in parts of the northwest and south-eastern coastal areas, where they exceed 4.5 million USD/km. The lowest-cost areas are dispersed across central, north-central, and north-eastern China. By 2030 (**Figure 1b**) and 2035 (**Figure 1c**), a substantial reduction in costs is observed across numerous regions because of technological advancements, infrastructure maturation, and achieved economies of scale. Concurrently, a significant expansion of red-shaded zones, particularly throughout northern and eastern China, signals the progressive exhaustion of accessible storage capacity in these regions. This trajectory persists through 2040–2050 (**Figure 1d-f**), as the saturation of optimal storage sites forces operations to shift toward locations with higher marginal costs. The minimum full-chain costs for DSA are 838 in 2025, 727 by 2030, and 347 by 2050 USD/km (see **Supplementary Table 7–8**).

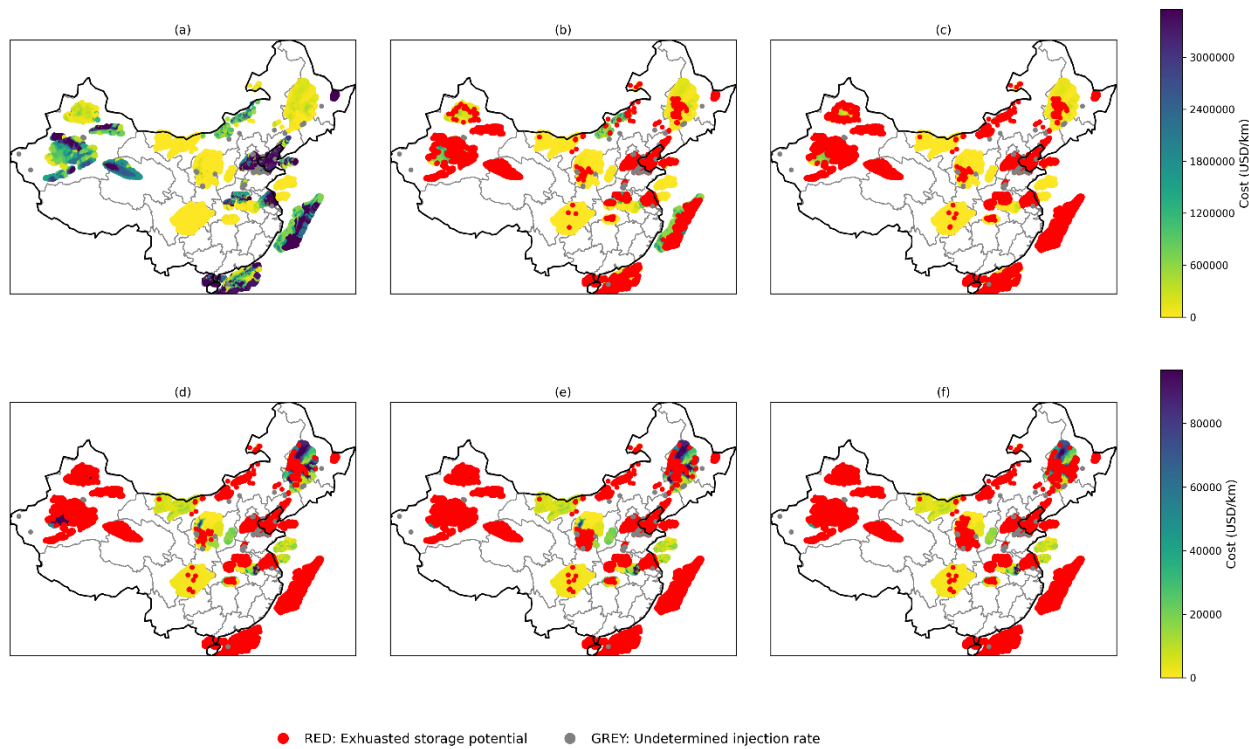


Figure 1: The average carbon sequestration cost for both DSA and EOR using pipeline transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The grey-shaded regions represent areas where storage potential remains untapped due to undetermined injection rates at those specific grid points, while the red-shaded areas indicate areas where the storage potential has been exhausted/fully utilized.

While the spatial cost patterns of tankers (Figure 2) are similar to those for pipelines (Figure 1), tanker costs are systematically higher, a difference that is most pronounced in the initial years (2025-2035) (Supplementary Table 8 and 10). The minimum costs for EOR are 518 in 2025, 1540 by 2030, and 736 by 2050 USD/km (see Supplementary Table 9 and 10).

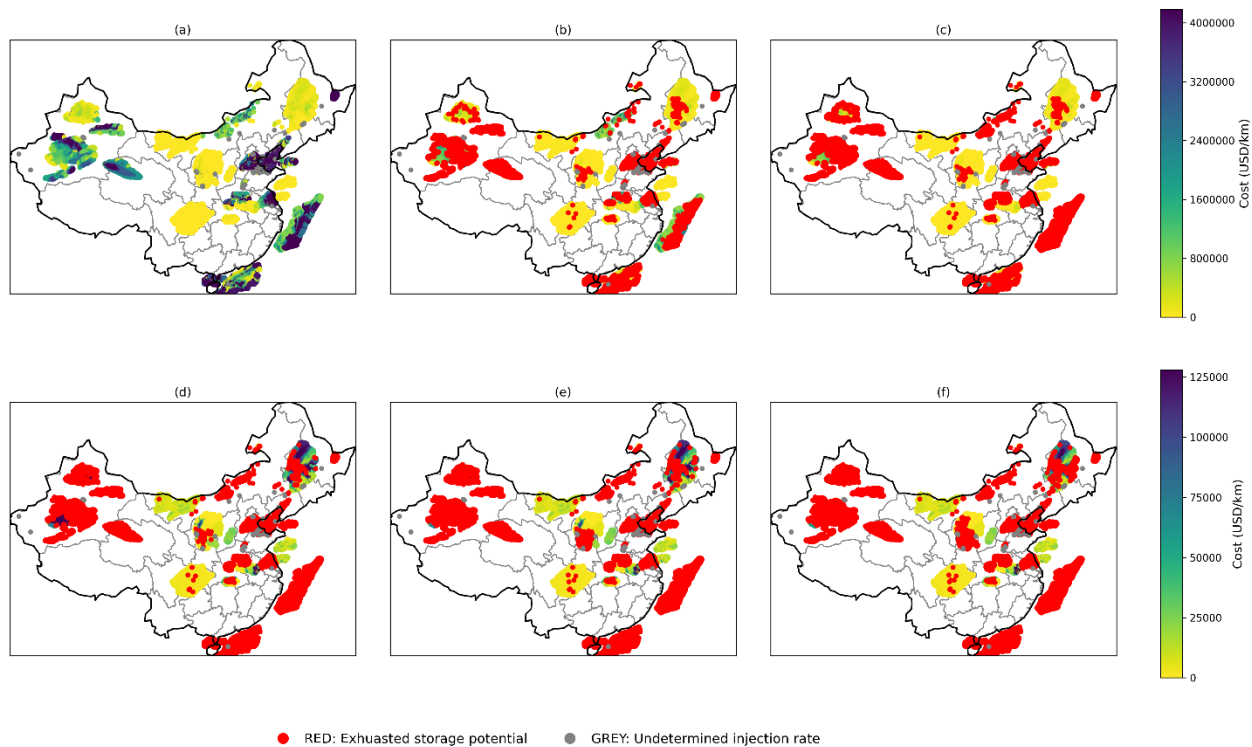


Figure 2: The average carbon sequestration cost for both DSA and EOR using tank transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The colour scheme is the same as explained in Figure 1.

Due to increased per-unit cost and logistical constraints associated with tanker-based CO₂ transport (Supplementary Figures 11 and 12), pipeline transport is more economical, yielding an average cost reduction of 14% relative to tanker transport across all DSA grid points in 2025. This cost advantage increases to 16% in 2030 and 29% by 2050. For EOR, the corresponding reductions are 14% (2025), 19% (2030), and 29% (2050) (Figure 3). These results highlight the economic importance of prioritizing pipeline infrastructure for long-term, scalable CCS deployment, especially in regions possessing high injection rates and substantial storage potential. The expansion of red-shaded regions, indicating exhausted storage capacity, occurs more rapidly in EOR scenarios than in DSA across all modelled scenarios (Supplementary Figure 5–16). This trend reflects the more limited distribution and lower overall storage potential of EOR sites compared to the extensive and capacious DSA (Supplementary Figure 17). Analysis across all scenarios reveals that the most favourable regions can sequester approximately 18, 8, and 4 MtCO₂ at 5th percentile costs across all regions of 2346 (2025), 1754 (2030), and 303 (2050) USD/km, respectively (Supplementary Figure 5–16). Similarly, the 10th percentile costs are 3048 (2025), 2198 (2030), and 832 (2050) USD/km, facilitating the sequestration of 52, 26, and 10 MtCO₂, respectively (Supplementary Table 7–10).

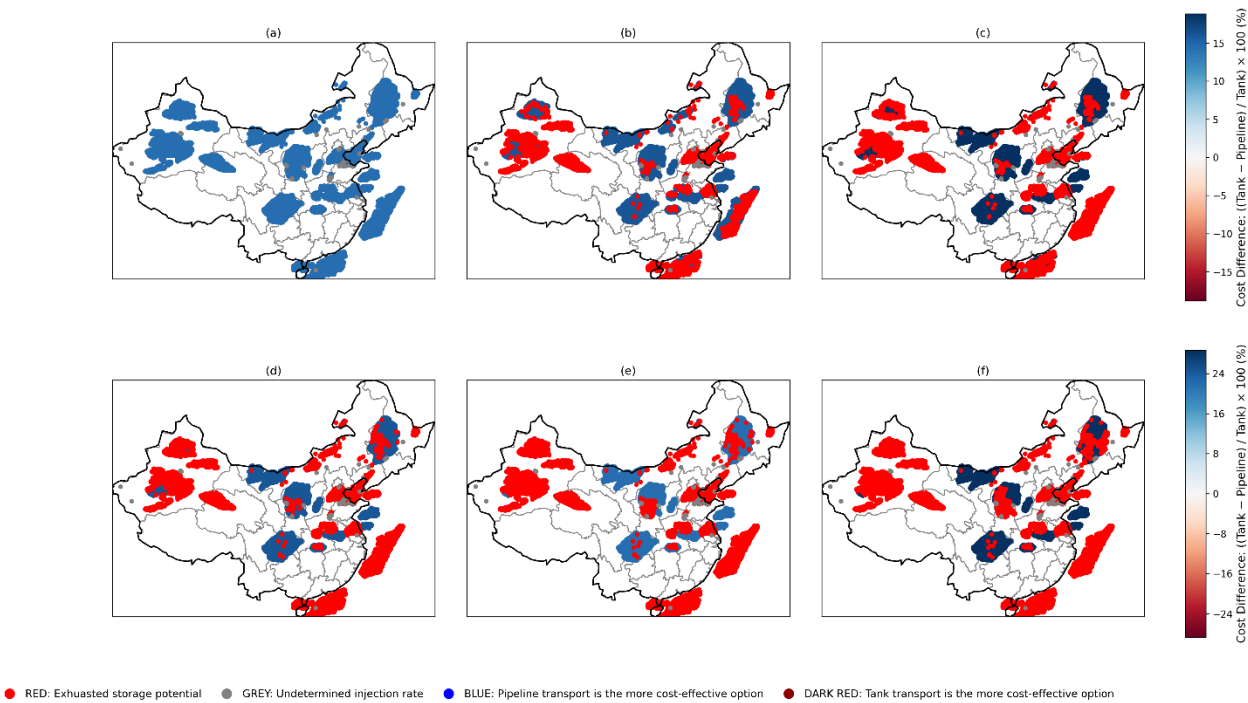


Figure 3: Illustration showing the areas where pipeline and tank transport are more economical for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050.

Risk assessment of CO₂ storage leakage

Active fault lines and associated seismicity pose substantial risks to the security of geological CO₂ storage. Injecting CO₂ into subsurface formations increases pore pressure, which can induce earthquakes³⁸. Seismic activity can create new fractures or reactivate existing faults and caprock seals, providing pathways for the stored CO₂ to migrate out of the deep reservoir and potentially reach the surface or contaminate freshwater aquifers³⁹. These risks warrant a detailed site-specific geotechnical evaluation before project deployment. Consequently, a thorough spatial analysis of the relationship between fault networks and prospective CO₂ storage reservoirs is essential for robust risk assessment and the selection of secure sequestration sites. Spatial analysis of geological CO₂ storage sites in China reveals significant heterogeneity in associated risk factors (Figure 4a).

Areas with very high-risk factors (>9) are located in the eastern and south-eastern coastal regions, coinciding with the identified DSA storage formations and accounting for 1.52 Gt of storage potential. This higher risk directly correlates with proximity to active tectonic faults and a historical frequency of seismic events (see Supplementary Figure 2 and 18), making stringent monitoring and proactive pressure management strategies advisable in these tectonically complex areas.

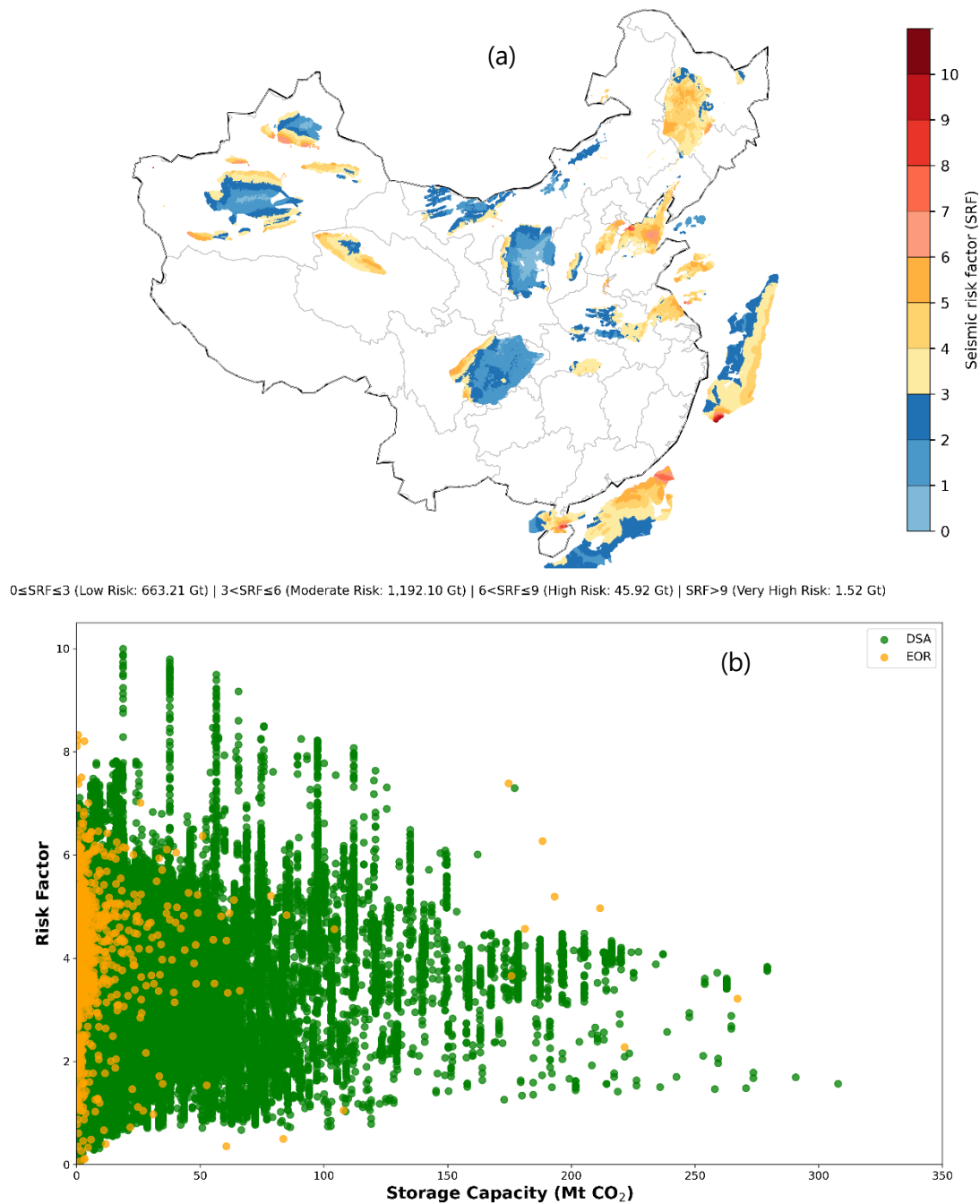


Figure 4: (a) Seismic risk factor spatial distribution and (b) risk factor at the DSA and EOR storage capacity.

Moderate to high risk (3-9) is observed across large parts of northern, eastern, and coastal regions of China, reflecting moderate seismic hazard influenced by regional faults and localized site amplification. In contrast, lower-risk sites (below 3) are primarily located in the north-western and central regions, such as Xinjiang and the Sichuan Basin. The relationship between storage potential

and risk is further depicted in [Figure 4b](#), which illustrates a general trade-off. Sites with higher storage capacity typically exhibit a higher risk, forming a major diagonal cluster. However, a cohort of sites, exclusively DSAs located in the lower-right quadrant, deviates from this trend by offering high storage capacity ($>0.2 \text{ GtCO}_2$) alongside a comparatively low risk (<3), identifying them as priority candidates for future storage projects. This geospatial analysis shows the importance of integrating fault and seismic risk data into site selection processes, identifying areas where CCS deployment would necessitate additional monitoring and mitigation strategies to ensure safe, reliable, and long-term containment.

The sensitivity analysis in [Supplementary Figure 19](#) highlights seismic risk factor as highly responsive to changes in the seismic hazard level, as defined by different Peak Ground Acceleration (PGA) exceedance probabilities. Compared to the base case (10% probability level of exceedance in 50 years (PEin50) (475-year return period)), the seismic risk factor does not remain constant but shows a systematic increase across some sites, with the magnitude of increase directly tied to the severity of the hazard scenario. The amplification risk increases up to 100% for 1% PEin50 (4,975-year return period), and 34% for 0.5% PEin50 (10,000-year return period), particularly accentuating risk in already high-hazard zones. Conversely, the most common shaking scenario (63% PEin50 (50-year return period)) results in the smallest changes, with risk factors increasing up to 14%, indicating a more uniform, lower-risk landscape due to frequent low-intensity shaking, which induces minimal amplification due to elastic soil behaviour. The risk factor increases up to 25% and 15% for the 2% (2,475-year return period) and 5% (975-year return period) probability exceedance levels. However, in localized regions, the risk factor can also decrease, with reductions of up to 38%, 131%, 16%, 7%, and 94% observed for the 0.5%, 1%, 2%, 5%, and 63% probability levels of exceedance in 50 years, respectively. This localized attenuation is observed exclusively on hard rock substrates or under conditions where the predominant period of seismic motion no longer coincides with the fundamental resonance frequency of the site⁴⁰. This systematic escalation confirms that the choice of hazard exceedance probability directly and substantially influences the final risk ranking, with the largest relative increase occurring for the lowest-probability (highest-intensity) scenarios, particularly in regions of already elevated seismic hazard.

[Supplementary Figure 20](#) reveals that the calculated seismic risk factor is significantly influenced by the methodological choice for estimating site amplification vulnerability, demonstrating that risk does not remain the same as the base case when this component is varied. Compared to the base case, which used the Boore et al.⁴¹ empirical model, the application of both the GB 50011-2010 code method⁴² and the assumption of no amplification results in risk factor increases of up to 100% across the storage site locations. The spatial pattern and approximate magnitude of increase are notably similar for both alternative methods, indicating a consistent directional shift in risk assessment when deviating from the empirically derived amplification model. This sensitivity highlights the site response component as a major source of epistemic uncertainty in the risk framework, where

conservative code-based classifications or the omission of amplification can substantially elevate the perceived risk, particularly at sites underlain by soft soils that are assigned higher vulnerability scores.

Supplementary Figure 21 demonstrates that the seismic risk factor is highly sensitive to the choice of the fault influence decay constant (λ), with both the magnitude and spatial pattern of risk changing substantially from the base case ($\lambda = 200$ km). Compared to the base case, the risk factor increases by up to 100% for $\lambda = 50$ km, up to 27% for $\lambda = 100$ km, up to 31% for $\lambda = 300$ km, and up to 50% for $\lambda = 500$ km. In other areas, the risk factor decreases by up to 1268% for $\lambda = 50$ km, 159% for $\lambda = 100$ km, 61% for $\lambda = 300$ km, and 820% for $\lambda = 500$ km. These divergent responses highlight that the fault proximity model's spatial influence is not uniform. This is a direct and dominant consequence of China's position at the confluence of several tectonic plates^{43,44}. This same dynamic geological environment underpins the marked spatial heterogeneity in seismic risk factors across the country. Hazard levels vary substantially because plate boundary forces exert regional influence, tectonic stress localises along discrete fault networks⁴⁵, and the underlying crustal blocks exhibit differing capacities for strain accumulation and release^{46,47}. A smaller λ sharply localizes risk to the immediate fault zones, drastically reducing risk elsewhere but massively amplifying it for sites very near faults, while a larger λ spreads the fault influence more diffusely, moderating extreme local highs but elevating risk over broader regions farther from the fault traces.

Integrating CO₂ storage sinks with industrial clusters

Effective CO₂ source-sink matching is a foundational step for deploying CCUS clusters and for designing associated pipeline networks, aiming to minimize transport costs and maximize carbon sequestration³³. Figure 5a presents the spatial distribution of manufacturing industries (CO₂ sources) and geological storage sinks at high resolution. Industrial activity is heavily concentrated in the eastern, central, and southern regions, which benefit from proximity to both DSA and EOR storage resources, creating substantial potential for source-sink integration. Among the total 8,697,995 manufacturing industrial sources, 89% lie within 250 km of a viable storage site, delineating optimal zones for pipeline infrastructure that would reduce both economic and environmental costs. Priority for CCUS deployment is given to the MC (metallurgical, chemical, and resource rough processing) industrial category (**Supplementary Figure 3**), a hard-to-abate sector facing significant barriers to electrification or renewable-based decarbonization, which contributes 31% of China's current total emissions⁴⁸. With China's total geological CO₂ storage potential estimated at over 1,000 GtCO₂, the country is well endowed with long-term cumulative storage capacity. However the average annual injection rate of 0.550 GtCO₂/year indicates that throughput is a practical bottleneck, meaning that storage deployment may still lag behind the pace required for decarbonization. The eastern and south-eastern regions present optimal conditions for integrated CCUS networks, hosting dense industrial clusters near geological storage. The spatial co-location of multiple industrial types in these areas improves the economic viability through shared infrastructure and economies of scale.

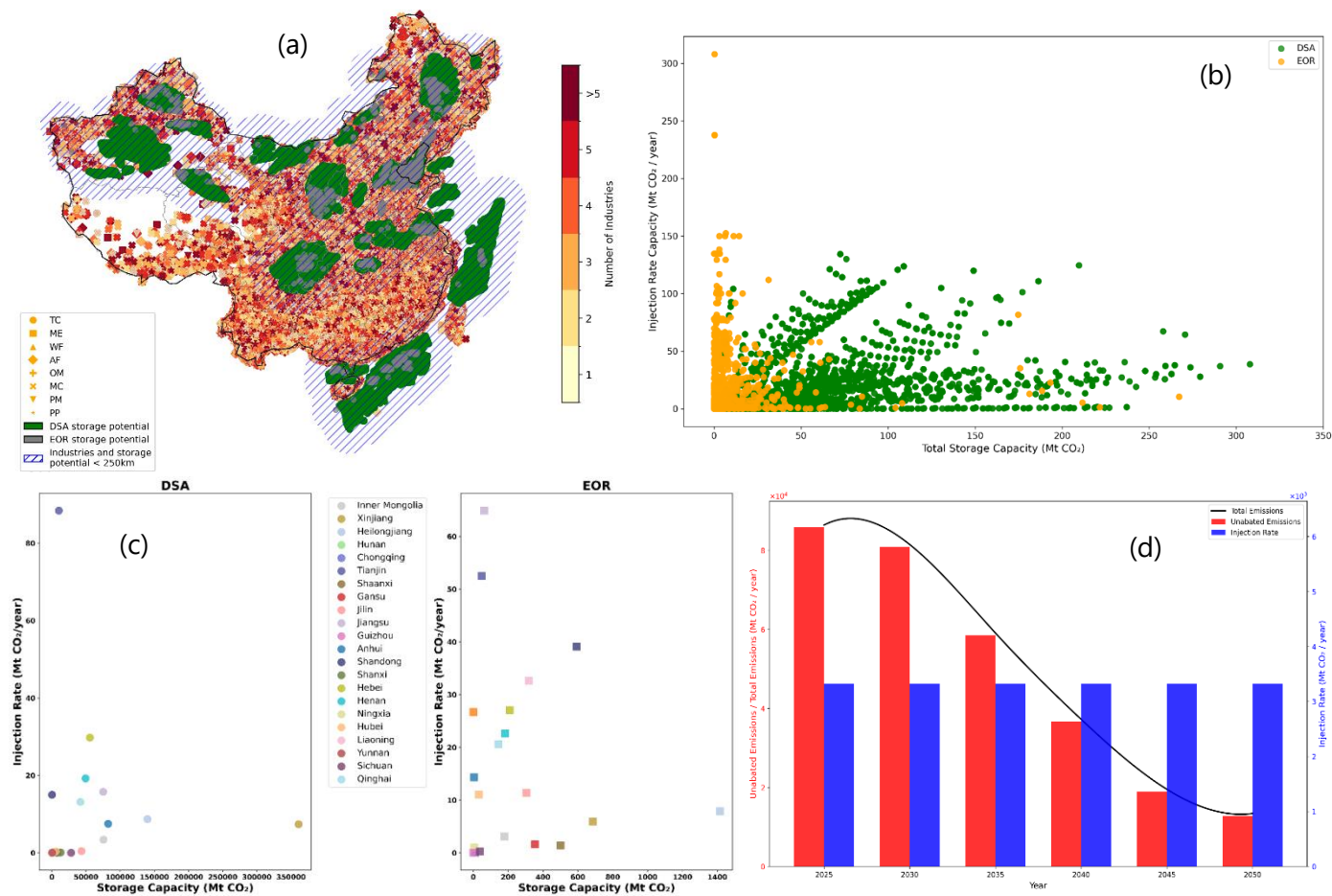


Figure 5: (a) The optimal location for synergistic alignment between CO₂ sources (industrial clusters) and CO₂ storage sinks. (b) Utilization plot at the 5x5 km grid level and (c) utilization plot at the provincial level. (d) injection rate vs. emission rate timeline. The spatial analysis reveals distinct patterns of industrial cluster proximity to geological storage sites. The purple-hatched shaded regions denote industrial facilities situated within a 250 km radius of potential storage locations, representing optimal candidates for CCS implementation using pipeline transportation infrastructure. TC: Textile and garment; ME: mechatronics and equipment; WF: wood furniture; AF: Food processing firms of agricultural and sideline products; MC: metallurgical, chemical, and resource rough processing; PM: pharmaceutical; PP: paper making; and OM: other manufacturing firms represent the classification of manufacturing industries.

Analysis at the provincial level shows a divergence in geological CO₂ storage resource characteristics (Figure 5c). DSAs have substantial storage potential capacity but slower injection rates, typified by provinces such as Xinjiang and Heilongjiang, which possess total capacities exceeding 100 Gt yet relatively modest injection rates (below 0.02 GtCO₂/year). This profile positions them as strategic, long-term storage assets. EOR sites in several provinces, including Tianjin, Jiangsu, Liaoning, and Shandong, exhibit a “small but fast” profile, with more constrained total storage (below 800 Mt) but

significantly higher injection rate capacities per unit of storage (above 0.03 GtCO₂/year). This makes them particularly suitable for rapid, near-term deployment to mitigate concentrated industrial emissions. Grid-level analysis confirms that this functional heterogeneity exists within provinces (Figure 5b). This observed duality implies that a two-pronged strategy is necessary: exploiting “small but fast” EOR resources for early, high-impact projects, while concurrently developing the vast “big but slow” DSA resources for the century-scale sequestration required to meet long-term climate objectives.

Figure 5d highlights the challenges confronting China’s pathway to carbon neutrality by 2060. Although the nation’s total CO₂ emissions are forecast to decline substantially, from over 86 GtCO₂/year in 2025 to approximately 13.4 GtCO₂/year by 2050⁴⁹, a critical bottleneck is evident. The rate of geological CO₂ injection capacity remains insufficient to manage the emission stream, resulting in a large and persistent gap of unabated CO₂ that endures through to 2050. Consequently, even with falling total emissions, the volume of CO₂ that cannot be captured and stored remains high, ~13 GtCO₂/year by mid-century. This analysis suggests that, under current storage rates, China’s carbon management strategy is not yet aligned with the net-zero trajectory. Net-zero CO₂ emissions are achieved when anthropogenic sources of CO₂ are balanced by an equivalent volume of removals through carbon sinks. Achieving the 2060 goal necessitates a parallel and accelerated investment in the development of storage infrastructure, clean energy deployment, and energy efficiency to bridge this gap. Without such integrated efforts, the nation will lack the necessary deployment rate capacity to neutralize its hard-to-abate sectoral emissions, thereby jeopardizing the feasibility of its long-term target.

We use Sankey diagrams to illustrate China’s decarbonization pathways for hard-to-abate sectors by tracking the flow of captured emissions to geological storage sites from 2025 to 2050. The analysis reveals a two-tiered storage strategy. Provinces such as Jiangsu, Shandong, and Hebei act as integrated hubs, sequestering a substantial share of their emissions locally through co-located storage resources, as visualized by circular markers and internal flow pathways (Figure 6a). In contrast, hard-to-abate clusters in provinces with limited storage potential require the development of cross-provincial transport networks to channel captured CO₂ to sink-rich provinces, including Xinjiang, Chongqing, and Qinghai (Figure 6b).

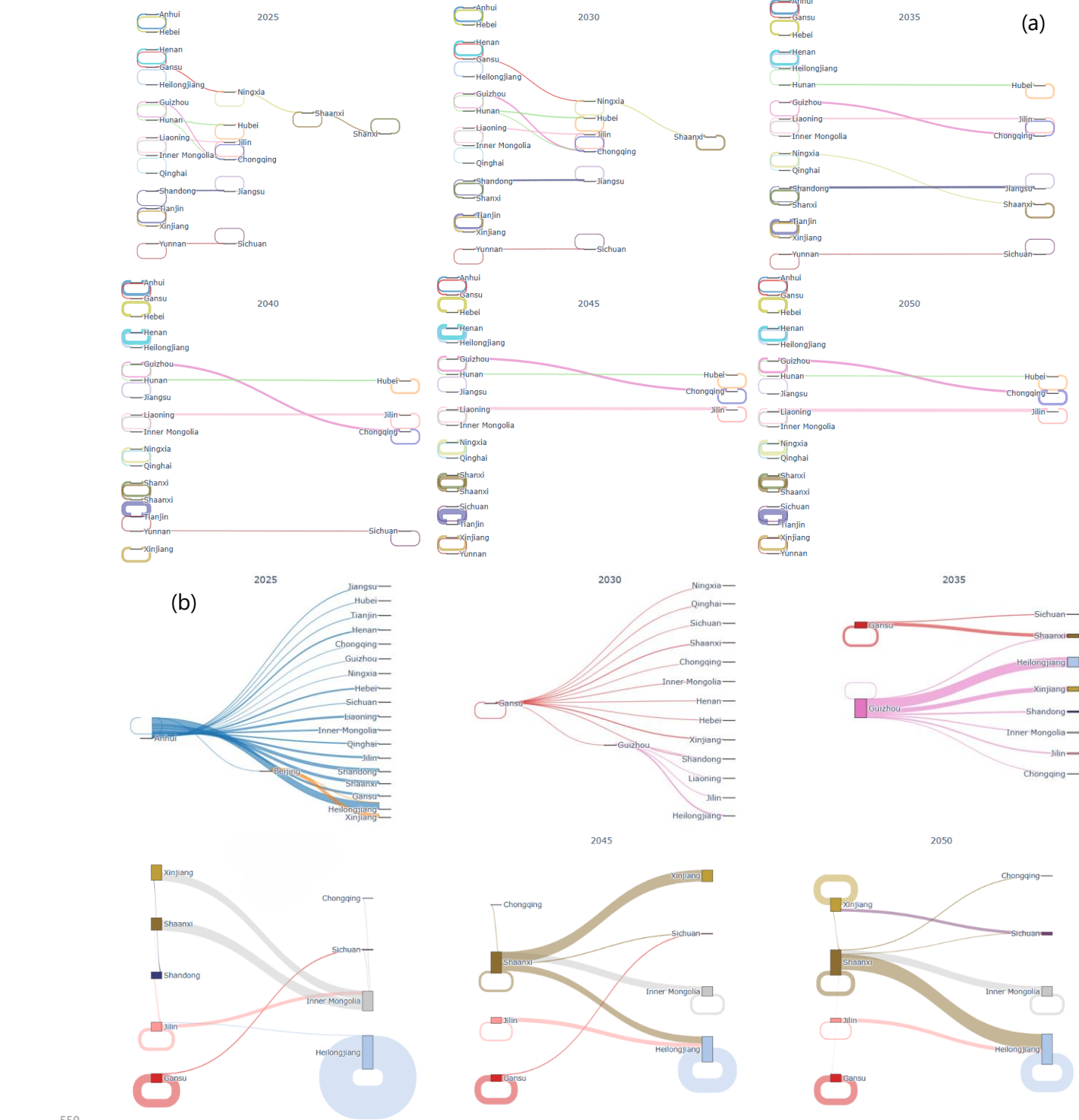


Figure 6: A Sankey diagram showing the flow of CO₂ from source sectors/provinces to (a) DSA and (b) EOR sink provinces for the MC manufacturing category. The width of the flows in (a) and (b) represents the average annual injection capacity rate of CO₂ (Mt/yr) (see [Supplementary Table 6](#)). Provinces with a circular marker act as self-contained "source-and-sink" hubs, utilizing their own storage capacity first, while linear flows indicate provincial cross-boundary transport of CO₂.

The limited capacity of EOR sites is rapidly depleted, shifting the long-term storage burden toward the more abundant yet slower-injecting DSA. These findings highlight the critical need for strategic pipeline infrastructure to connect major hard-to-abate industrial centres with extensive DSA basins, thereby ensuring a permanent waste disposal route essential for meeting China's climate objectives.

Spatial coupling of blue hydrogen production and CO₂ storage

Hydrogen has been proposed as a decarbonization substitute for fossil fuels in heavy industries⁵⁰. Despite China's status as the world's largest hydrogen producer, 99% of its current output is grey hydrogen, from natural gas via steam methane reforming (SMR), while green hydrogen contributes less than 1%⁵¹. The integration of hydrogen into the future energy system faces a significant cost barrier, ranging from 3.20 to 8 USD/kg in 2024⁵². Given that green hydrogen, produced through water electrolysis powered by renewable energy, remains comparatively expensive^{53,54}, blue hydrogen from natural gas via SMR with CCS presents a cost-competitive alternative to conventional fossil fuels. In regions of China where fossil fuel is abundant, CO₂ storage is accessible, and renewable energy supply is limited, the production of blue hydrogen is likely to be a cost-effective option⁵⁵. The large-scale adoption of blue hydrogen is contingent on the minimization of associated CCS facility costs and the maximization of capture efficiency⁵⁶, as incomplete CO₂ removal undermines the climate benefits that distinguish blue hydrogen from its unabated fossil-based counterpart.

Figure 7 identifies optimal areas across China for the co-location of blue hydrogen production facilities with geological CO₂ storage reservoirs, highlighted by purple-hatched shading where most of China's natural gas fields and storage potential fall within a 250 km radius. This strategic proximity is highly advantageous, as it substantially reduces the requirement for long-distance CO₂ transportation, a primary cost component within the CCS value chain. The most promising regions include north-eastern China, the Ordos Basin, central and eastern Sichuan, and specific areas within Xinjiang in the northwest, where extensive natural gas reserves overlap with substantial CO₂ storage capacity. Several southern coastal provinces demonstrate strong potential for co-location, supported by dense clusters of natural gas fields and proximal DSA storage resources. By integrating blue hydrogen production with local CCS infrastructure, these areas can achieve considerable cost advantages.

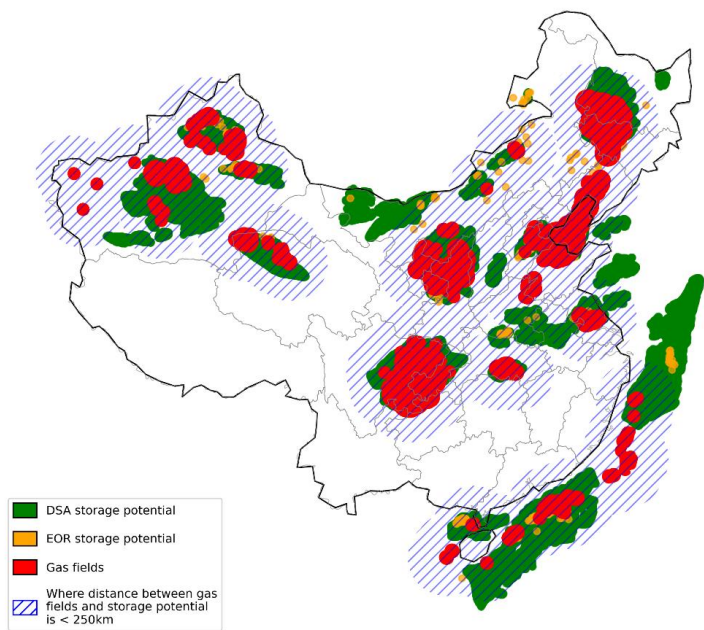


Figure 7: The optimal location for blue-hydrogen facilities. The purple-hatched shaded region shows areas located near natural gas fields and CO₂ geological storage sites within a 250 km radius.

Based on the priority regions identified, we estimate a potential for China to establish 15 to 25 major blue hydrogen production hubs by 2050 (Table 1), wherein multiple emission sources would utilize shared transport and storage infrastructure. For example, a region with substantial resources, such as the Ordos Basin, given its extensive natural gas reserves and coincident storage potential, could support 3–5 major facilities (0.3–1.0 Mt H₂/year total), whereas smaller regions might accommodate 1–2 (0.1–0.4 Mt H₂/year total). The aggregate annual blue hydrogen production capacity from this national network of hubs is projected to range from 2.5 to 3.0 Mt H₂/year by 2050, corresponding to approximately 15 to 25 major hubs, Assuming 0.10–0.20 Mt H₂/year per hub, consistent with large-scale SMR-CCS facility benchmarks^{57,58}. A sensitivity analysis around key model parameters sees this result range from 0.8 Mt H₂/year (4–8 hubs) under slow deployment, to 4.5 Mt H₂/year (22–45 hubs) under accelerated deployment.

Provincial contribution to emission reduction

Supplementary Figure 22a presents the projected contribution of DSA-based CCS to provincial emission reductions between 2025 and 2050, which is the cumulative amount of CO₂ sequestered by a particular year relative to emissions in that year for each province in percentage. Provinces like Jiangsu (295%), Qinghai (126%), Tianjin (141%), Shandong (165%), and Hebei (105%) store more than they produce, effectively importing emissions from other provinces. They provide the largest contributions to sequestering about 4.4 GtCO₂ by 2050, a pattern attributed to their substantial geological storage potential and comparatively high injection rate capacities. Many coastal and less

geologically endowed provinces, such as Beijing, Fujian, Guangdong, Hainan, Jiangxi, Shanghai, Tibet, and Zhejiang, have no CCS contributions throughout the period, reflecting the absence of DSA-based storage capacity (Supplementary Table 11). EOR-based CCS (Supplementary Figure 22b) demonstrates strong performance in Shandong (260%), Jilin (167%), Qinghai (55%), and Jiangsu (48%) provinces by 2050, sequestering about 1.1 Gt of CO₂, owing to the presence of mature oil fields that offer synergistic opportunities for combining CO₂ storage with enhanced oil recovery (Supplementary Table 12). EOR offers near-term emission reduction potential in hydrocarbon-rich regions, while DSA provides large-scale, long-term storage capacity in sedimentary basins, highlighting the complementary roles of EOR and DSA in China's decarbonization strategy.

Figure 8a analyses the DSA temporal distribution from 2025 to 2050, showing an upward shift of provincial data points as national emissions decline, while the inherent storage capacity remains fixed. Provinces including Xinjiang (with a storage-to-emissions ratio increasing from 99 in 2025 to 558 by 2050), Jiangsu (from 24 to 559), Jilin (from 26 to 255), Heilongjiang (from 45 to 309), Inner Mongolia (from 30 to 192), and Qinghai (from 32 to 158) persistently plot well above the parity line, indicating a consistent and robust storage surplus (see Supplementary Table 13). In contrast, highly industrialized eastern provinces such as Shandong (ratio declining from 0.04 in 2025 to 0 by 2040), Guizhou, Liaoning, and Hunan (all 0 for all years) remain below this line (see Supplementary Table 13), signalling a persistent storage deficit and a growing imperative to transport CO₂ to storage-endowed regions via pipelines or tank transportation. Figure 8b presents the corresponding analysis for EOR sites. In 2025, most provinces appear above the parity line, reflecting the early economic attractiveness and deployment of EOR, particularly in regions with active oil fields like Xinjiang (ratio of 0.2) and Heilongjiang (0.5) (see Supplementary Table 13). A distinct trend is seen: the number of provinces represented steadily decreases from over 20 in 2025 to just 8 by 2050 as their finite EOR capacity is fully allocated (see Supplementary Table 4). By 2050, the EOR landscape is considerably constrained, with only a few provinces, such as Xinjiang (0.8), Gansu (1.2), and Heilongjiang (2.7), maintaining a viable storage-to-emissions balance. This trajectory demonstrates that while EOR offers a near-term storage pathway, its limited capacity renders it insufficient for supporting China's large-scale, long-term carbon management strategy.

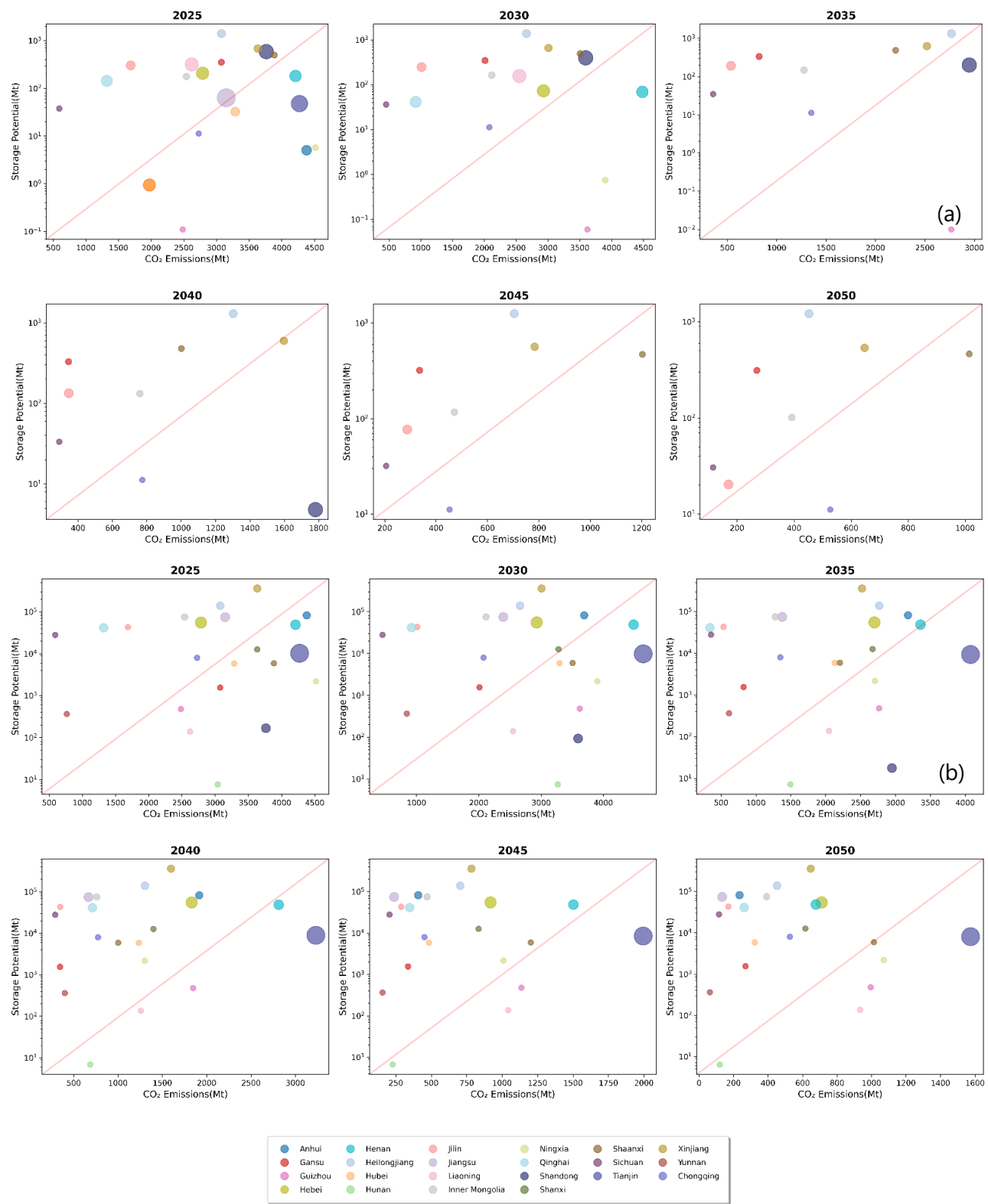


Figure 8: The responsibility vs. capability of each of the provinces for (a) DSA and (b) EOR. Each bubble represents a province, with its size indicating the average annual injection rate. The red diagonal line ($y = x$) serves as a benchmark: provinces above the line are “storage-rich”, while those below the line are “emission-rich”.

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Discussion

This analysis projects a dramatic decline in China's full-chain CCS pipeline costs. Costs are projected to fall from 2025 to 2050 as technologies mature, and with economies of scale. The economic advantage of pipeline transport highlights supports the development of shared pipeline infrastructure, particularly within optimal source-sink matching zones. This aligns with IEA assessments that CCUS hubs with collective CO₂ transport and storage networks can accelerate deployment⁶⁰. The energy transition, which reduces fossil fuels in final energy consumption, presents an opportunity to repurpose existing oilfield pipelines for CO₂ transport. Repurposing offers 40–80% lower investment cost and 20–60% shorter construction times relative to new-build construction^{61,62}. Decommissioned pipelines where rights-of-way remain available also present promising opportunities for repurposing⁶³. A nationally coordinated CO₂ pipeline network, analogous to the natural gas grid, could connect the eastern industrial centers with western and northern storage basins. EOR sites provide a near-term storage opportunity, creating a window to plan and develop the more extensive DSA resources required to meet long-term climate mitigation objectives⁶⁴.

Figure 9 shows China's competitive advantage in unit CCS costs relative to Europe and the United States (see [Supplementary Table 14](#)). Disaggregated cost components show that China's capture costs (32–43 USD/tonne) are comparable to the lower end of the US range (25–30 USD/tonne) and well below the EU range. The advantage is greater for transport and storage, with China's pipeline transport (0.11–0.19 USD/tonne.km) and storage costs (7–8 USD/tonne) being a fraction of the combined costs reported in Western studies (25–35 USD/tonne in the US and 8–40 USD/tonne in the EU). This gap may reflect lower labour and construction costs and the shorter planned distances for source-sink matching within our optimized model. However, this cost advantage exists within a very different policy context, as the US and EU benefit from more mature regulatory frameworks and stronger financial drivers, such as the US 45Q tax credit⁶⁵ and the high EU Emissions Trading System (ETS) carbon price^{66,67}. These can exceed total CCS cost and greatly reduce investment risk. Lower costs in China present an opportunity, but large-scale deployment depends on implementing similarly robust carbon pricing or fiscal incentives to bridge the remaining economic gap and attract capital.

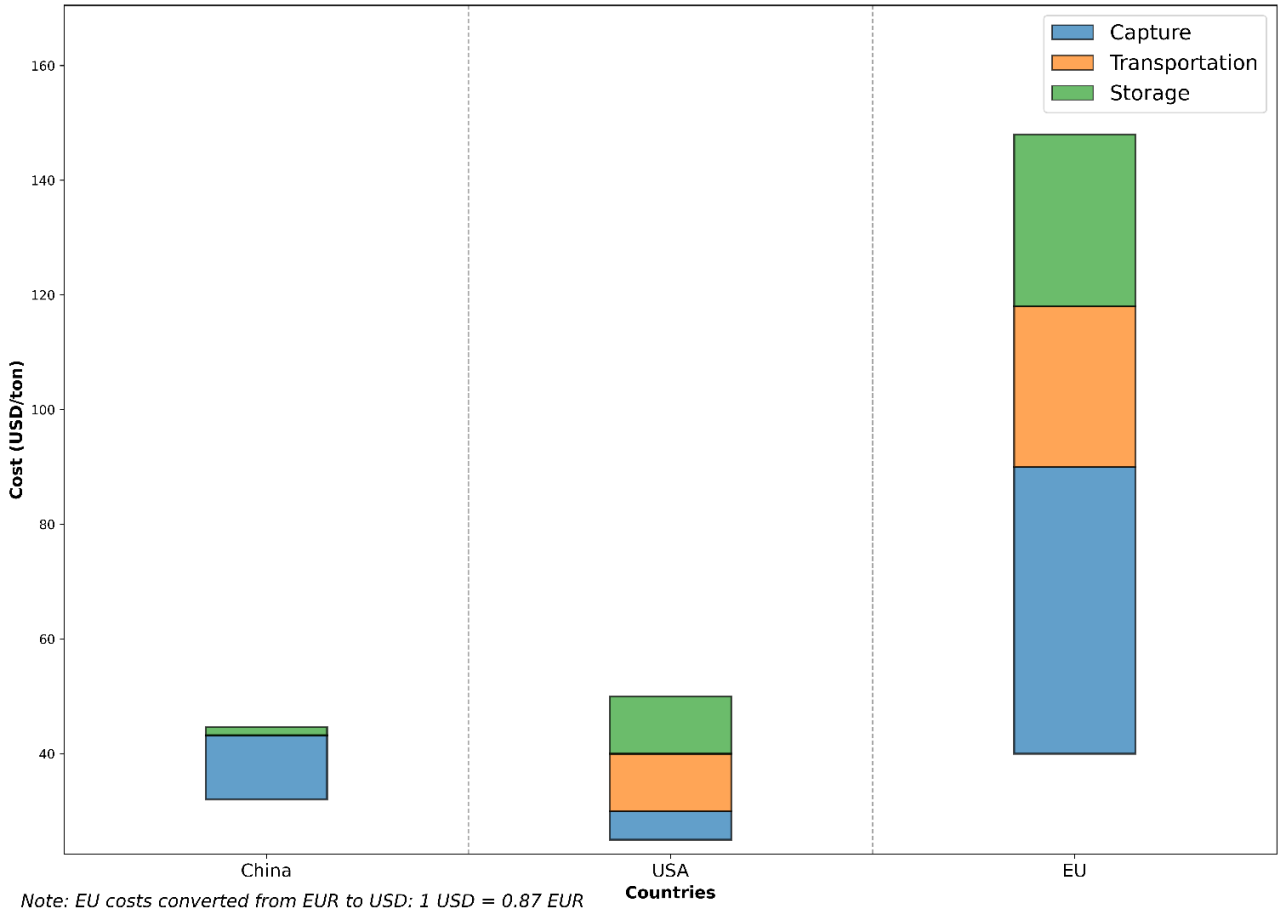


Figure 9: Comparison of CCS unit costs with reference to China^{59,68–70}.

Deployment should prioritize integrated hubs, such as Jiangsu and Shandong, where dense hard-to-abate industrial clusters co-exist with nearby geological storage. These regions offer the lowest-cost decarbonization, enabling rapid deployment with minimal transportation and infrastructure. However, Figure 6 shows that local solutions are insufficient nationally. A larger fraction of emissions, particularly from storage-constrained provinces, must cross provincial borders. This creates a coordination challenge that requires strategic design of a national CO₂ pipeline network linking emission hotspots to sink-rich regions. This long-term infrastructure must target a clear end-state of a shift to DSA. While EOR provides a valuable “small-but-fast” bridge for early projects, its capacity will be rapidly exhausted. Long-term decarbonization of China's hard-to-abate sectors therefore depends on timely development of “big-but-slow” DSA resources, as the only storage option with sufficient scale to support the national 2060 carbon neutrality goal.

Our integrated risk assessment highlights a key geomechanical constraint for China's carbon management strategy. Expansive storage formations in the North China Plain and Xinjiang face increased risks of CO₂ leakage and seismicity due to nearby active faultlines, requiring a split strategy for site development. Initial efforts should focus on high-capacity, low-risk DSA sites that can support early, large-scale deployment. Demonstrating safe operation in geologically stable areas will support

creating regulatory frameworks and build public trust. Higher-risk, high-capacity reservoirs, needed to meet national storage targets, require strict management. Their development must rely on strong monitoring, verification, and accounting (MVA) systems linked to financial assurance methods to handle long-term liability and remediation. This dual approach maintains storage security across geologically different locations while preserving the size of deployment required to satisfy climate targets.

To accelerate CCS deployment, policymakers should support early-stage CCUS development and demonstration through targeted subsidies, tax credits, or differentiated carbon pricing that reflects varying storage costs. Higher-cost DSA sites should receive greater incentives than lower-cost EOR fields, encouraging development across the full cost spectrum rather than concentrating projects in favourable locations. Otherwise, near-term market forces risk triggering a “gold rush” on EOR due to its economic returns, potentially diverting investment and delaying essential DSA capacity. Strategic designation of CCS Hub Priority Development Zones at the national level could optimize infrastructure through integrated CO₂ trunk pipelines, shared compression facilities, and unified permitting processes. Provincial governments should incorporate CCS capacity targets into their five-year climate plans, aligned with regional geological constraints and validated injection rate limits.

Specialized hydrogen-CCS industrial parks should be prioritized in regions where natural gas resources, CO₂ storage, and industrial clusters are co-located, supported by expedited approval for integrated facilities. A risk-based regulatory framework should provide streamlined approvals for low-risk sites, and strengthened requirements for higher-risk locations. A dynamically updated CCS deployment mapping system should be developed, with revisions made every 3-5 years to incorporate new geological surveys, industrial activity data, and infrastructure. This would ensure site selection reflects the relevant economic and risk considerations. China’s export industries should be incentivized to adopt CCS to comply with emerging carbon border adjustment mechanisms and international green supply chain requirements, strengthening the positioning of domestic manufacturers in low-carbon global markets.

Conclusions

CCS is a key pathway for China to achieve carbon neutrality. Although CCS can decarbonize hard-to-abate sectors, its deployment is constrained by cost that vary with CO₂ concentration and project scale⁵⁹. This study assesses CO₂ storage costs, source-sink matching, and provincial contributions to emission reduction, using a 5 km resolution national dataset of onshore and offshore storage potential and injection capacity across 24 sedimentary basins and 1,181 oilfields.

We identify optimal regions where costs can decline by almost an order of magnitude by 2050, sequestering 3–18 MtCO₂/year. Even cost-effective storage must be close to emissions sources to deliver low full-chain costs. Our geospatial ranking identifies optimal source-sink clusters, particularly

for industrial emissions in south-eastern and eastern China. Co-location strategies reduce CO₂ transport costs and improve the scalability of blue hydrogen as a cost-competitive alternative to fossil fuels.

The analysis proposes a dual storage strategy in which “small-but-fast” enhanced oil recovery sites are given preference for near-term decarbonization, while “big-but-slow” deep saline aquifers are established for supporting long-term climate targets. It also highlights provincial disparities, separating “storage-rich” regions such as Xinjiang and Inner Mongolia from “emission-rich” provinces including Shandong and Guizhou. Inter-provincial cooperation towards coordinated national CCS infrastructure is therefore needed to allow emission-rich provinces to offset industrial emissions by accessing lower-cost storage elsewhere.

While constrained by uncertainties inherent in subsurface modelling and simplified economic indices, this study establishes a roadmap for developing China’s CCS industry. Future work should develop a spatially explicit inventory with separate exclusion layers to enable more nuanced evaluations. We also recommend work to incorporate dynamic plume modelling, high-resolution infrastructure routing, and probabilistic risk frameworks.

China’s commitment to become carbon neutral by 2060 faces an implementation gap, since the theoretical CCS potential lacks a strategic framework for cost-effective, safe, large-scale deployment. Existing evaluations often lack the spatial resolution and scope needed to guide investment. This study uses a high-resolution geospatial dataset to identify CO₂ storage sites and determine effective CCS implementation pathways to meet national climate targets. Without spatially detailed planning, CCS might remain a symbolic pledge rather than a deployable solution.

End matter

Data availability

Results from the model that support the findings of this study are presented in the article and its Supplementary Information. All underlying figure data is provided in the GitHub code repository (<https://github.com/Caroline-Acen/China-CCS-Project>).

Code availability

The model used for this analysis is available from the GitHub code repository in Python and Excel (<https://github.com/Caroline-Acen/China-CCS-Project>).

Credit author contributions

Caroline Acen: Conceptualization, Data curation, Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Olusola Bamisile:** Writing – original draft, Writing – review & editing, Validation, Supervision, Investigation, Methodology. **Daniel O. Olasehinde:** Writing – original draft, Data curation, Formal analysis, Visualization. **Joeri Regelj:** Investigation, Supervision, Validation, Writing – review & editing. **Omosalewa O. Olagundoye:** Data curation, Writing – original draft. **Chukwuebuka J. Ejiyi:** Supervision, Writing – review & editing. **Qi Huang:** Supervision, Resources, Project administration, Funding acquisition. **Iain Staffell:** Investigation, Supervision, Validation, Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Results from the model that support the findings of this study are presented in the article and its Supplementary Information. All underlying figure data is provided in the GitHub code repository (<https://github.com/Caroline-Acen/China-CCS-Project>).

Supplementary Material

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Appendix 1: Literature Review

China has committed to peak carbon emissions before 2030 and reach carbon neutrality by 2060, setting one of the most ambitious climate goals worldwide [1]. To achieve its climate goals, the government recognizes that, in addition to expanding renewable energy and improving energy efficiency, carbon capture, utilization, and storage (CCUS) will be essential. This is particularly important for hard-to-abate sectors like power, chemicals, steel, and cement, which are key to China's economy but also some of the most challenging to decarbonize [2]. As of 2020, China's CCS efforts have resulted in the capture and storage of around 0.5 million to 2 million tons of CO₂, with large-scale projects underway to reduce emissions further. The integration of CCS technologies into industries like cement, steel, and power generation is a growing element of achieving carbon abatement targets aligned with the nation's carbon-neutrality goals [3]. As of 2025, more than 100 CCUS projects are either operational or under development, with over 40 projects having capture capacities exceeding 100,000 tonnes of CO₂ [4]. Still, large-scale deployment has not matched expectations, due to high costs, uneven source–sink distribution, and limited policy incentives. Nevertheless, the 14th Five-Year Plan places CCUS among the country's priority technologies, signaling stronger support for scaling up in the coming decades [3].

Against this policy backdrop, recent studies have explored China's carbon sequestration potential from different perspectives the study by Ma et al., [5] report that from 1985 to 2022, Sanjiangyuan National Park sequestered a total of 96.6 million tons of carbon, with the economic value of the carbon stock increasing from 93.51 million yuan in 1985 to 5.78 billion yuan in 2022. Geologic carbon storage possesses significant technical potential, with existing estimates ranging from 10,000 to 40,000 GtCO₂ [18], [19]. At a national scale, it is estimated that China's onshore basins alone could store 448 GtCO₂, a volume sufficient to permanently sequester the country's cumulative industrial emissions from 1990 to 2060 [20]. While these aggregate figures suggest ample capacity, realizing this potential requires detailed planning to connect emission sources with suitable storage sites. One approach to this challenge involves global-scale matching of carbon sources and sinks. A recent cost-effectiveness analysis identified 3,093 carbon clusters and 432 sinks across 85 countries and regions, projecting a total mitigation of 92 GtCO₂ through CCUS [21]. Of this, 64% would be sequestered in sedimentary basins for aquifer storage, with the remainder utilized for CO₂-enhanced oil recovery (EOR). However, the practical implementation of such a globally optimized scheme remains contingent upon unprecedented levels of international cooperation in both finance and technology transfer, limiting its immediate applicability for national planning [21].

Narrowing the focus to China, an offshore source-sink matching analysis estimated the storage capacity of the country's offshore sedimentary basins at 767.3 Gt, with saline aquifers contributing the vast majority (764.3 Gt) and oil and gas reservoirs providing the remainder [14]. This work highlighted the immense potential of offshore storage but did not fully integrate it with onshore emission patterns. Building on this, a subsequent study developed a source-sink matching model that

explicitly incorporated onshore–offshore geospatial connectivity to optimize realistic pipeline networks for China. This high-resolution analysis estimated the multisectoral CCUS supply potential from coal-fired power, steel, cement, and coal chemical industries at approximately 1.75, 0.77, 0.56, and 0.23 GtCO₂ per year, respectively [15]. Despite its advancements, the model faced challenges in identifying the most suitable plants for retrofitting, and its reliance on economic and technical parameters not yet validated by large-scale operational projects introduced uncertainties into the optimal layout [15]. Addressing some of these uncertainties, another line of inquiry linked spatially explicit source-sink matching with bottom-up energy planning to propose multi-sector shared CCUS networks incorporating plant-level industrial transfer and infrastructure reuse. This analysis projected that by 2050, a total annual capture of 1.74 GtCO₂ would require nearly 19,000 km of CO₂ trunk lines, with the power sector contributing the largest share (1.09 Gt), followed by cement (0.42 Gt), iron and steel (0.19 Gt), and ammonia industries (0.04 Gt) [17]. Nevertheless, the study did not account for secondary modes of CO₂ transport within individual emission hubs, and it similarly grappled with the limited availability of techno-economic data from currently operational large-scale projects.

A more sector-specific approach integrated transformation pathways for China's steel industry through 2060, simulating the distribution of steel units and generating optimal CCS deployment schemes. The findings indicated that CCS could account for 31.4–40.7% of carbon mitigation in the steel sector by 2060, achieving emission reductions of 472.4–609.6 Mt at levelized costs ranging from 187.4 to 193.5 Chinese Yuan per tCO₂ [22]. The model also projected that over 650 steel units would require either elimination or retrofitting to align with sectoral decarbonization goals [22]. However, this analysis did not extend to environmental and social dimensions, such as material consumption, land use changes, or job creation, which are critical for holistic policy assessment. Recognizing the inherent uncertainties in long-term planning, another study employed fuzzy possibility and interval linear programming to optimize CCS cluster development under multiple uncertainties, including policy shifts, economic fluctuations, technological change, and storage prospects. The results demonstrated that cluster-based approaches could reduce pipeline lengths by approximately 81.5–81.8% compared to point-to-point projects and lower plant-level levelized costs by 35.7–86.9%, achieving savings of 13.8–49.0 USD per tCO₂ [16]. Yet, the design and deployment of CO₂ transport networks remain sensitive to regional factors such as geographic conditions and population density, which were not fully integrated into the optimization framework [16].

A common thread running through the aforementioned studies is the limited attention paid to risk assessment at geological storage sites themselves. A risk-based planetary assessment has suggested a more prudent global storage limit of approximately 1,460 GtCO₂, with roughly 70% located onshore and 30% offshore, underscoring the need to account for geological uncertainty [23]. Storing carbon in geological formations can potentially reactivate existing active faults, leading to induced seismicity and subsequent CO₂ leakage, hazards that could undermine both environmental integrity and public acceptance. An investigation into these risks at a potential storage site in the southern San Joaquin Basin, California, revealed that leakage through faults is strongly dependent on fault depth, aperture,

and relative permeability [24]. The study concluded that further characterization of fault properties is essential for predicting leakage rates with the accuracy required for safe site selection. This finding highlights a critical gap in existing CCUS planning frameworks for China, which have yet to integrate quantitative geomechanical risk assessment into high-resolution source-sink matching and infrastructure optimization.

Appendix 2: Challenges to the large-scale deployment of CCS in China

One of the most critical barriers to the large-scale deployment of CCS in China is the high abatement cost. Although CCUS in China costs about 120–480 RMB/tCO₂ (\$20–80/tCO₂), mostly from capture (60–80%), and although lower than global averages, it still raises coal power, steel, and cement costs significantly [21]. Retrofitting coal-fired power plants for CCS substantially increases costs, and in the cement sector, the estimated cost of CO₂ avoidance is about US\$70/tCO₂ under 2012 assumptions, with economic viability highly sensitive to carbon price growth, making CCS retrofits in cement plants less attractive compared to coal power [22]. These significant cost burdens reduce the competitiveness of CCS relative to alternative low-carbon technologies, discouraging large-scale private investment and restricting the technology's potential for expansion.

Another major challenge lies in the spatial mismatch between CO₂ sources and sinks. China has a vast CO₂ storage potential of about 1,004,319 Mt in deep saline aquifers, 21,287 Mt in depleted oil reservoirs, and 15,321 Mt in depleted gas fields, with 76 cities able to store local coal plant emissions for over 10 years, yet many major industrial hubs in the east remain distant from these sites, creating costly source–sink mismatches [23]. This raises transportation costs, requires extensive pipeline infrastructure, and is further complicated by limited site characterization. Similar issues are evident in the LNG sector, where carbon emissions increased by 2.5% in 2024 and CCS projects face repeated failures, including injection well integrity breaches, CO₂ transport instabilities, and financial overruns, highlighting the broader risks of scaling CCS without integrated spatial and technical planning [24].

The urgency of CCS deployment further complicates China's pathway, as global key industries such as thermal power, iron and steel, and cement contribute over 50% of anthropogenic CO₂ emissions, making CCUS indispensable, yet cost-effective plant-level strategies remain underexplored with plant-by-plant optimization indicates that strategic deployment could lower cumulative costs by 18.5 USD/t by 2030, with about 20% of China's older facilities (>20 years) showing unexpected low-cost potential due to proximity to EOR storage sites, though viability remains highly sensitive to oil prices [25]. Decommissioning them prematurely would create severe stranded asset risks, but delaying CCS rollout could jeopardize the carbon neutrality target. At the global level, momentum is accelerating with over 570 CCUS projects in development and 368 expected to be operational by 2030 with a combined capacity of 743 Mt of CO₂ per year, still falling short of net-zero pathways and prompting stronger competition in the United States, United Kingdom, European Union, and Canada, where CCUS is coupled with nature-based solutions including the planting of 2 billion trees by 2030 [26].

China's CCUS sector suffers from a lack of stable commercial models, with most projects still at pilot or lab scale despite the country's strong R&D progress and industrial chain maturity, as commercialization is projected only for 2030–2035 with an optimal deployment potential of 248.54–564.90 GW, while costs for second-generation capture technologies are expected to fall from 219

CNY/tCO₂ to 165 CNY/tCO₂ and annual expenditures peak at 23.92 billion CNY (corporate) in 2035 and 63.71 billion CNY (government subsidies) in 2044 [27]. Coal-to-hydrogen enterprises and oil fields in China are presently unwilling to engage in CCUS projects without stronger incentives, but empirical evidence shows that targeted policies such as a blue hydrogen price of CNY 35.6/kg H₂, a carbon market price of CNY 150/t CO₂, or a carbon tax of about CNY 130/t CO₂ combined with subsidies can effectively reduce investment risks, align stakeholder interests, and promote sustainable private sector participation [28]. Without such structures, China faces difficulty in transforming demonstration efforts into large-scale commercial deployment.

Finally, insufficient policy incentives remain a major barrier, as China lacks commercial-scale CCS demonstration projects and direct incentive mechanisms, whereas simulations show that adopting policies like the ETS and 45Q tax credit could enable 960–1,604 MtCO₂ annual reductions by 2060, cut ETS carbon prices by 71.2–82.7%, and raise GDP by 5.64–6.59%, benefits already illustrated by the EU's inclusion of CCS in its ETS since 2009 and the U.S. 45Q tax credit [10]. This uncertainty undermines investor confidence and slows project execution, making it essential for China to strengthen financial incentives and regulatory clarity while integrating CCS into its national carbon market, which as of January 2024 encompassed 28 global trading systems covering 17% of global emissions and one-third of the world's population, so that CCS can assume a central role in achieving the country's carbon neutrality strategy [29].

Supplementary Tables

Supplementary Table 1: The economic cost of each component of the full chain CCS system [33], [34], [35].

Year		2025	2030	2035	2040	2045	2050
Capture Cost (CNY/ton)	Pre-combustion	100-180	90-130	70-80	50-70	30-50	20-40
	Post-combustion	230-310	190-280	160-220	100-180	80-150	70-120
	Oxy-fuel combustion	300-480	160-390	130-320	110-230	90-150	80-130
Transportation (CNY/ton.km)	Tank	0.9-1.4	0.8-1.3	0.7-1.2	0.6-1.1	0.5-1.1	0.5-1.0
	Pipeline	0.8	0.7	0.6	0.5	0.5	0.4
Storage cost (CNY/ton)		50-60	40-50	35-40	30-35	25-30	20-25

Supplementary Table 2: Minimum, maximum, and average costs of each segment of the full chain CCS system.

Year	Post-Combustion Capture (CNY/ton)	Tank Transport (CNY/ton.km)	Pipeline Transport (CNY/ton.km)	Storage (CNY/ton)
	Min / Avg / Max	Min / Avg / Max	Fixed	Min / Avg / Max
2025	230 / 270 / 310	0.9 / 1.15 / 1.4	0.8	50 / 55 / 60
2030	190 / 235 / 280	0.8 / 1.05 / 1.3	0.7	40 / 45 / 50
2035	160 / 190 / 220	0.7 / 0.95 / 1.2	0.6	35 / 37.5 / 40
2040	100 / 110 / 180	0.6 / 0.85 / 1.1	0.5	30 / 32.5 / 35
2045	80 / 115 / 150	0.5 / 0.8 / 1.1	0.5	25 / 27.5 / 30
2050	70 / 95 / 120	0.5 / 0.75 / 1.0	0.4	20 / 22.5 / 25

Supplementary Table 3: CO₂ transport technology [33], [35].

Distance	Type of pipeline	Expected Implementation Timeline
< 250 km	Gaseous	2025
Greater than 250 km	Supercritical	2030
1000 – 4000 km	High-pressure supercritical CO ₂	2050

Supplementary Table 4: Remaining storage potential for DSA and EOR.

Provinces	DSA (Mt)					EOR (Mt)				
	2030	2035	2040	2045	2050	2030	2035	2040	2045	2050
Anhui	82329.31	82291.86	82254.41	82216.96	82179.51	0	0	0	0	0
Beijing	0	0	0	0	0	0	0	0	0	0
Fujian	0	0	0	0	0	0	0	0	0	0
Gansu	1557.45	1556.85	1556.25	1555.65	1555.05	345.18	336.98	328.78	320.58	312.38
Guangdong	0	0	0	0	0	0	0	0	0	0
Guangxi	0	0	0	0	0	0	0	0	0	0
Guizhou	479.67	479.62	479.57	479.52	479.47	0.06	0.01	0	0	0
Hainan	0	0	0	0	0	0	0	0	0	0
Hebei	55522.26	55373.36	55224.46	55075.56	54926.66	73.31	0	0	0	0
Henan	49141.11	49044.96	48948.81	48852.66	48756.51	69.13	0	0	0	0
Heilongjiang	139759.26	139715.61	139671.96	139628.31	139584.66	1375.47	1335.87	1296.27	1256.67	1217.07
Hubei	5873.47	5871.72	5869.97	5868.22	5866.47	0	0	0	0	0
Hunan	7.39	7.14	6.89	6.64	6.39	0	0	0	0	0
Jilin	43323.49	43321.34	43319.19	43317.04	43314.89	247.34	190.59	133.84	77.09	20.34
Jiangsu	75010.62	74931.82	74853.02	74774.22	74695.42	0	0	0	0	0
Jiangxi	0	0	0	0	0	0	0	0	0	0
Liaoning	138.53	138.03	137.53	137.03	136.53	155.49	0	0	0	0
Inner Mongolia	75442.43	75425.48	75408.53	75391.58	75374.63	163.04	147.59	132.14	116.69	101.24
Ningxia	2184.96	2184.16	2183.36	2182.56	2181.76	0.75	0	0	0	0
Qinghai	41669.7	41603.85	41538	41472.15	41406.3	41.3	0	0	0	0
Shandong	92.69	17.69	0	0	0	396.11	200.46	4.81	0	0
Shanxi	12715.15	12714.6	12714.05	12713.5	12712.95	0	0	0	0	0
Shaanxi	5949.02	5948.52	5948.02	5947.52	5947.02	493.27	486.17	479.07	471.97	464.87
Shanghai	0	0	0	0	0	0	0	0	0	0
Sichuan	28086.27	28086.22	28086.17	28086.12	28086.07	36.3	34.85	33.4	31.95	30.5
Tianjin	9856.89	9414.84	8972.79	8530.74	8088.69	0	0	0	0	0
Tibet	0	0	0	0	0	0	0	0	0	0
Xinjiang	361155.31	361118.41	361081.51	361044.61	361007.71	656.13	626.53	596.93	567.33	537.73
Yunnan	364.5	364.4	364.3	364.2	364.1	0	0	0	0	0
Zhejiang	0	0	0	0	0	0	0	0	0	0
Chongqing	8024.67	8024.62	8024.57	8024.52	8024.47	11.31	11.26	11.21	11.16	11.11

Supplementary Table 5: Classification of the manufacturing firms in China [36].

Classification of the manufacturing industry	Industry	ObjectID	Grid cell number
Textile and garment	Textile industry Textile clothing, apparel industry Leather, fur, feather, and their products, and footwear	TC15, TC19	Number of textile and garment firms in 2015 and 2019
Mechatronics and equipment	General equipment manufacturing Electrical machinery and equipment manufacturing Special equipment manufacturing Manufacturing of computers, communications, and other electronic equipment Automobile industry Manufacturing of railway, marine, aerospace, and other transportation equipment Instrument manufacturing Repair of metal products, machinery, and equipment	ME15, ME19	Number of mechatronics and equipment firms in 2015 and 2019
Papermaking culture	Culture and education, art, sports, and recreation goods manufacturing Paper and paper products industry Printing and recording media reproduction	PP15, PP19	Number of papermaking culture firms in 2015 and 2019
Wood furniture	Wood processing and wood, bamboo, rattan, palm, grass products Furniture manufacturing	WF15, WF19	Number of wood furniture firms in 2015 and 2019
Pharmaceutical manufacturing	Pharmaceutical manufacturing	PM15, PM19	Number of pharmaceutical manufacturing firms in 2015 and 2019
Agricultural, sideline products, and food processing	Agricultural and sideline food processing industry Food manufacturing Wine, beverage, and refined tea manufacturing Tobacco products industry	AF15, AF19	Number of food processing firms of agricultural and sideline products in 2015 and 2019
Other manufacturing	Other manufacturing	OM15, OM19	Number of other manufacturing firms in 2015 and 2019
Metallurgical chemical industry and resource rough processing	Chemical raw materials and chemical products manufacturing Metal products industry Rubber and plastic products Ferrous metal smelting and rolling processing industry Nonferrous metal smelting and rolling processing industry Petroleum processing, coking, and nuclear fuel processing industries Chemical fiber manufacturing Comprehensive utilization of waste resources Nonmetallic mineral products industry	MC15, MC19	Number of metallurgical, chemical, and resource rough processing firms in 2015 and 2019

Supplementary Table 6: Provincial-level carbon geological storage potential and injection rate for DSA and EOR [30].

Province name	Storage potential- DSA (Mt)	Storage potential- EOR (Mt)	Injection rate capability-DSA (Mt/a) (Average)	Injection rate capability-EOR (Mt/a) (Average)	CO ₂ emissions					
					2025	2030	2035	2040	2045	2050
Anhui	82366.76	5.04	7.49	14.3	4378.58	3689.193	3178.903	1914.627	407.6902	234.5317
Beijing	0	0.94	0	26.66	1971.05	1749.212	1599.127	1016.713	516.4738	304.3205
Fujian	0	0	0	0	2457.08	2572.074	1816.407	852.1229	357.342	201.0201
Gansu	1558.05	353.38	0.12	1.64	3074.12	2009.136	821.882	344.5652	335.3974	269.3226
Guangdong	0	0	0	0	2348.14	2291.625	1376.898	888.6936	312.2568	123.547
Guangxi	0	0	0	0	2476.31	2897.593	2146.759	1597.467	542.2746	210.9802
Guizhou	479.72	0.11	0.01	0.01	2482.21	3619.174	2765.459	1845.836	1135.347	995.2192
Hainan	0	0	0	0	963.445	498.3007	276.9303	150.038	135.2558	113.2484
Hebei	55671.16	208.36	29.78	27.01	2785.88	2927.749	2697.633	1829.404	917.5485	711.3168
Henan	49237.26	182.33	19.23	22.64	4209.61	4483.425	3357.883	2811.342	1501.845	676.964
Heilongjiang	139802.91	1415.07	8.73	7.92	3075.4	2660.751	2767.858	1302.179	703.9461	452.2365
Hubei	5875.22	32.49	0.35	11.01	3285.78	3294.182	2130.112	1234.053	483.3584	323.119
Hunan	7.64	0	0.05	0	3037.34	3265.892	1493.915	685.0753	226.6651	120.6104
Jilin	43325.64	304.09	0.43	11.35	1685.28	1012.49	536.3834	346.1764	287.368	170.0831
Jiangsu	75089.42	63.6	15.76	64.89	3148.21	2392.055	1375.431	665.7734	237.5608	133.5603
Jiangxi	0	0	0	0	4825.6	4954.436	3220.602	2135.195	1146.049	279.8916
Liaoning	139.03	318.79	0.1	32.66	2620.37	2548.185	2047.436	1255.681	1042.518	933.9088
Inner Mongolia	75459.38	178.49	3.39	3.09	2538.88	2114.86	1275.748	759.0119	470.5361	391.8857
Ningxia Hui	2185.76	5.75	0.16	1	4514.6	3900.996	2704.848	1300.292	1008.396	1070.795
Qinghai	41735.55	144.4	13.17	20.62	1320.22	918.3045	339.9303	711.7122	346.997	261.9921
Shandong	167.69	591.76	15	39.13	3762.54	3592.19	2948.12	1780.689	574.745	226.8584
Shanxi	12715.7	0	0.11	0	3631.23	3277.331	2671.737	1398.884	833.0377	616.472
Shaanxi	5949.52	500.37	0.1	1.42	3884.05	3505.478	2203.683	1000.838	1202.234	1014.024
Shanghai	0	0	0	0	2948.21	3689.671	2890.601	2061.6	700.621	342.0936
Sichuan	28086.32	37.75	0.01	0.29	590.588	454.0111	356.1728	290.127	205.2565	115.8563
Taiwan	0	0	0	0	4269.97	4636.808	4075.412	3229.798	1994.145	1573.513
Tianjin	10298.94	48.05	88.41	52.51	373.818	267.6379	162.4412	187.6462	99.86054	49.90242
Tibet	0	0	0	0	3630.85	3006.386	2519.751	1596.678	783.1039	647.1894
Xinjiang	361192.21	685.73	7.38	5.92	763.973	843.5587	613.3887	398.9311	155.8779	62.97559
Yunnan	364.6	0	0.02	0	2636.05	2336.831	1366.908	884.213	431.9158	265.3013
Zhejiang	0	0	0	0	2727.67	2075.529	1349.722	773.832	451.8649	526.8225
Chongqing	8024.72	11.36	0.01	0.01	4378.58	3689.193	3178.903	1914.627	407.6902	234.5317
Macao	0	0	0	0						
Hong Kong	0	0	0	0						
Total	999733.2	5087.86	209.81	344.08	86417.05	81485.06	59088.08	37249.19	19547.49	13419.56

Supplementary Table 7: Statistics of the DSA full chain cost using pipeline transport mode.

Year		2025	2030	2035	2040	2045	2050
Minimum (USD/km)	Minimum	766.61	646.83	551.00	407.26	367.33	303.45
	Mean	2132419.12	382949.73	107756.94	58376.26	31217.26	14786.78
	Maximum	35924139.74	8723530.35	3688730.12	1803467.38	485969.54	214060.14
	5th percentile	2345.81	1753.52	661.28	488.77	440.85	303.45
	10th percentile	3048.00	2198.49	1686.05	1236.84	1093.80	831.52
	95th percentile	9261527.12	1596294.26	615155.71	361653.39	129209.86	67923.45
Average (USD/Km)	Minimum	838.48	726.68	602.91	427.23	427.23	347.37
	Mean	2332333.41	430227.48	117907.96	61177.51	36287.53	16917.48
	Maximum	39292027.84	9800509.40	4036219.19	1891872.65	565203.71	245042.53
	5th percentile	2565.72	1970.00	723.57	512.73	512.73	347.37
	10th percentile	3333.75	2469.91	1844.88	1297.47	1272.14	951.88
	95th percentile	10129795.29	1793367.63	673105.16	379381.50	150276.69	77754.48
Maximum (USD/Km)	Minimum	910.35	806.54	654.81	543.02	487.12	391.29
	Mean	2531646.49	477261.66	127930.96	77758.33	41374.57	19056.47
	Maximum	42659915.94	10877488.46	4383708.26	2404623.18	644437.87	276024.92
	5th percentile	2785.64	2186.48	785.87	651.69	584.61	391.29
	10th percentile	3619.50	2741.33	2003.71	1649.12	1450.48	1072.23
	95th percentile	10998063.46	1990440.99	731054.61	482204.53	171343.52	87585.50

Supplementary Table 8: Statistics of the DSA full chain cost using tank transport mode.

Year		2025	2030	2035	2040	2045	2050
Minimum (USD/km)	Minimum	806.54	686.75	590.93	447.19	367.33	343.38
	Mean	2243482.61	406588.60	115565.42	64099.43	31217.26	16732.40
	Maximum	37795188.68	9262019.87	3956029.40	1980277.91	485969.54	242225.95
	5th percentile	2467.98	1861.76	709.20	536.69	440.85	343.38
	10th percentile	3206.75	2334.20	1808.22	1358.10	1093.80	940.94
	95th percentile	9743898.33	1694830.94	659732.21	397109.61	129209.86	76860.75
Average (USD/Km)	Minimum	978.23	866.43	742.65	566.97	547.01	487.12
	Mean	2721055.65	512963.53	145237.62	81268.91	46486.57	23736.67
	Maximum	45840699.15	11685222.75	4971766.68	2510709.50	723672.04	343622.85
	5th percentile	2993.35	2348.84	891.29	680.45	656.49	487.12
	10th percentile	3889.37	2944.89	2272.50	1721.87	1628.82	1334.82
	95th percentile	11818094.50	2138246.02	829122.91	503478.25	192410.34	109035.01
Maximum (USD/Km)	Minimum	1149.92	1046.10	894.38	782.58	726.68	630.86
	Mean	3197869.26	619022.55	174734.97	112063.48	61722.71	30723.70
	Maximum	53886209.61	14108425.62	5987503.96	3465486.35	961374.53	445019.76
	5th percentile	3518.71	2835.93	1073.38	939.21	872.12	630.86
	10th percentile	4571.99	3555.58	2736.77	2376.67	2163.83	1728.70
	95th percentile	13892290.68	2581661.09	998513.62	694941.82	255610.82	141209.28

Supplementary Table 9: Statistics of the EOR full chain cost using pipeline transport mode.

Year		2025	2030	2035	2040	2045	2050
Minimum (USD/km)	Minimum	473.92	1371.17	1168.04	863.33	778.69	643.27
	Mean	1032751.04	336712.17	232242.24	130726.34	125901.01	70544.56
	Maximum	21866709.48	5136920.03	2996258.98	1503532.52	1356127.37	563815.65
	5th percentile	8527.14	1542.57	1284.84	863.33	778.69	643.27
	10th percentile	24543.07	2315.27	1494.58	1037.97	876.03	723.67
	95th percentile	4197102.85	1201817.56	1023770.52	708596.72	670943.75	324378.39
Average (USD/Km)	Minimum	518.35	1540.46	1278.07	905.65	905.65	736.37
	Mean	1105278.12	335481.37	222616.02	117933.15	124071.99	70671.64
	Maximum	23916713.50	5771107.69	3278515.26	1577235.10	1577235.10	645420.54
	5th percentile	9302.18	1733.01	1397.89	905.65	905.65	736.37
	10th percentile	26812.87	2578.84	1635.37	1074.85	1018.86	828.42
	95th percentile	4586307.89	1321579.68	1049001.02	600617.32	699534.13	209586.68
Maximum (USD/Km)	Minimum	0.22	1709.74	1388.10	1151.11	1032.61	829.48
	Mean	847745.07	372347.45	241781.64	149896.34	141465.26	79607.14
	Maximum	25966717.51	6405295.35	3560771.54	2004710.03	1798342.82	727025.44
	5th percentile	15.49	1923.45	1518.24	1151.11	1032.61	829.48
	10th percentile	45.51	2861.56	17776.17	1366.17	1161.69	933.16
	95th percentile	3975073.15	1466808.22	1139312.37	763401.45	797599.66	236086.14

Supplementary Table 10: Statistics of the EOR full chain cost using tank transport mode.

Year		2025	2030	2035	2040	2045	2050
Minimum (USD/km)	Minimum	498.60	1455.81	1252.68	947.97	778.69	727.91
	Mean	1063172.29	317048.33	218193.18	123444.04	106678.72	69859.33
	Maximum	23005600.60	5454013.86	3213379.19	1650937.67	1356127.37	638001.92
	5th percentile	8947.81	1637.79	1370.12	947.97	778.69	727.91
	10th percentile	25791.43	2436.58	1602.88	1125.08	876.03	818.90
	95th percentile	4411591.40	1248965.42	1028159.94	628683.55	601468.60	207177.63
Average (USD/Km)	Minimum	604.74	1836.70	1574.31	1201.89	1159.57	1032.61
	Mean	1289491.14	319918.38	274215.76	156509.41	158858.53	99102.76
	Maximum	27902832.41	4711508.68	4038436.01	2093153.12	2019450.55	905072.48
	5th percentile	10852.55	2008.89	1721.90	1201.89	1159.57	1032.61
	10th percentile	31281.69	2350.17	2014.43	1426.44	1304.52	1161.69
	95th percentile	5350692.54	1507504.78	1292146.96	797080.93	895665.19	293903.16
Maximum (USD/Km)	Minimum	710.88	2217.58	1895.94	1658.95	1540.46	1337.32
	Mean	1515809.99	482945.71	330238.33	216027.08	211038.34	128346.20
	Maximum	32800064.22	8307858.32	4863492.83	2889140.93	2682773.72	1172143.05
	5th	12757.28	2494.77	2073.69	1658.95	1540.46	1337.32
	10th percentile	36771.94	3711.53	2425.98	1968.89	1733.01	1504.48
	95th percentile	6289793.68	1902493.83	1556133.97	1100196.21	1189861.79	380628.68

Supplementary Table 11: The contribution of each province's CCS to emission reduction for DSA.

Province	The percentage contribution					
	2025	2030	2035	2040	2045	2050
Anhui	0.1711	1.0151	2.3562	5.8680	36.7436	79.8400
Beijing	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Fujian	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Gansu	0.0039	0.0299	0.1460	0.5224	0.7156	1.1139
Guangdong	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Guangxi	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Guizhou	0.0004	0.0014	0.0036	0.0081	0.0176	0.0251
Hainan	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Hebei	1.0690	5.0858	11.0393	24.4178	64.9121	104.6650
Henan	0.4568	2.1446	5.7268	10.2602	25.6085	71.0156
Heilongjiang	0.2839	1.6405	3.1541	10.0562	24.8030	48.2601
Hubei	0.0107	0.0531	0.1643	0.4254	1.4482	2.7080
Hunan	0.0016	0.0077	0.0335	0.1095	0.4412	1.0364
Jilin	0.0255	0.2123	0.8017	1.8632	2.9927	6.3204
Jiangsu	0.5006	3.2942	11.4582	35.5076	132.6818	294.9978
Jiangxi	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Liaoning	0.0038	0.0196	0.0488	0.1195	0.1918	0.2677
Inner Mongolia	0.1335	0.8015	2.6573	6.6995	14.4091	21.6262
Ningxia	0.0035	0.0205	0.0592	0.1846	0.3173	0.3736
Qinghai	0.9976	7.1708	38.7432	27.7570	75.9084	125.6717
Shandong	0.3987	2.0879	5.0880	12.6356	52.1971	165.3014
Shanxi	0.0030	0.0168	0.0412	0.1180	0.2641	0.4461
Shaanxi	0.0026	0.0143	0.0454	0.1499	0.1664	0.2465
Shanghai	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Sichuan	0.0017	0.0110	0.0281	0.0517	0.0974	0.2158
Tianjin	2.0705	9.5335	21.6935	41.0598	88.6696	140.4660
Tibet	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Xinjiang	0.2033	1.2274	2.9289	6.9331	18.8481	28.5079
Yunnan	0.0026	0.0119	0.0326	0.0752	0.2566	0.7940
Zhejiang	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Chongqing	0.0004	0.0024	0.0074	0.0194	0.0443	0.0475

Supplementary Table 12: The contribution of each province's CCS to emission reduction for EOR.

Province	The percentage contribution					
	2025	2030	2035	2040	2045	2050
Anhui	0.1151	0.1366	0.1585	0.2632	1.2362	2.1490
Beijing	0.0477	0.0537	0.0588	0.0925	0.1820	0.3089
Fujian	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Gansu	0.0533	0.4081	1.9954	7.1394	9.7794	15.2234
Guangdong	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Guangxi	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Guizhou	0.0004	0.0014	0.0036	0.0060	0.0097	0.0111
Hainan	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Hebei	0.9695	4.6128	7.7238	11.3895	22.7083	29.2922
Henan	0.5378	2.5249	5.4299	6.4855	12.1404	26.9335
Heilongjiang	0.2575	1.4883	2.8614	9.1232	22.5017	43.7824
Hubei	0.3351	0.9863	1.5253	2.6328	6.7217	10.0551
Hunan	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Jilin	0.6735	5.6050	21.1602	49.1801	78.9928	166.8302
Jiangsu	2.0202	2.6588	4.6240	9.5528	26.7721	47.6189
Jiangxi	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Liaoning	1.2464	12.5105	15.5702	25.3878	30.5788	34.1350
Inner Mongolia	0.1217	0.7305	2.4221	6.1066	13.1340	19.7124
Ningxia	0.0222	0.1282	0.2126	0.4422	0.5702	0.5370
Qinghai	1.5619	11.2272	42.4793	20.2891	41.6142	55.1162
Shandong	1.0400	5.4465	13.2729	33.2321	102.9604	260.8499
Shanxi	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Shaanxi	0.0366	0.2025	0.6444	2.1282	1.7717	3.5009
Shanghai	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Sichuan	0.0491	0.3194	0.8142	1.4993	2.1193	6.2578
Tianjin	1.1253	1.0363	1.1790	1.4877	2.4096	3.0537
Tibet	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Xinjiang	0.1630	0.9846	2.3494	5.5615	15.1193	22.8681
Yunnan	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Zhejiang	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Chongqing	0.0004	0.0024	0.0074	0.0194	0.0443	0.0475

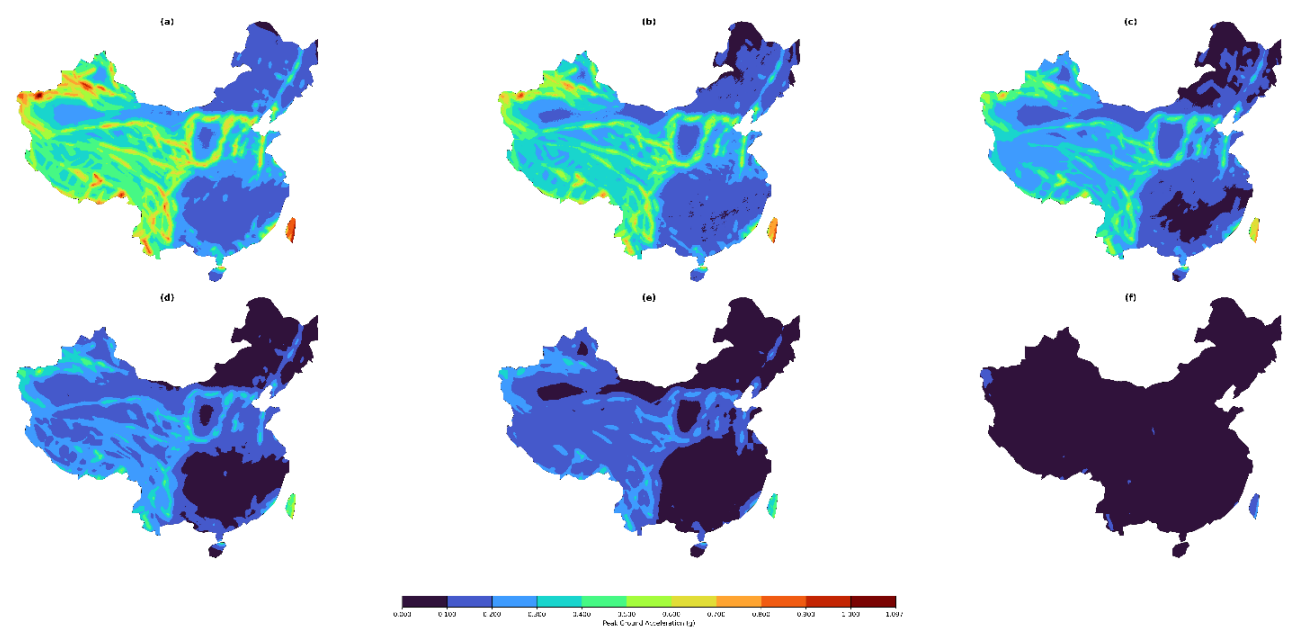
Supplementary Table 13: Storage-to-emissions ratio for DSA and EOR. Storage (Supplementary Table 4) divide by emissions (Supplementary Table 6).

Province	DSA						EOR					
	2025	2030	2035	2040	2045	2050	2025	2030	2035	2040	2045	2050
Anhui	19	22	26	43	202	350	0.001	0	0	0	0	0
Beijing	0	0	0	0	0	0	0.0005	0	0	0	0	0
Fujian	0	0	0	0	0	0	0	0	0	0	0	0
Gansu	1	1	2	5	5	6	0.1	0.2	0.4	1.0	1.0	1.2
Guangdong	0	0	0	0	0	0	0	0	0	0	0	0
Guangxi	0	0	0	0	0	0	0	0	0	0	0	0
Guizhou	0	0	0	0	0	0	0.00004	0.00002	0.000004	0	0	0
Hainan	0	0	0	0	0	0	0	0	0	0	0	0
Hebei	20	19	21	30	60	77	0.07	0.03	0	0	0	0
Henan	12	11	15	17	33	72	0.04	0.02	0	0	0	0
Heilongjiang	45	53	50	107	198	309	0.5	0.5	0.5	1.0	1.8	2.7
Hubei	2	2	3	5	12	18	0.01	0	0	0	0	0
Hunan	0	0	0	0	0	0	0	0	0	0	0	0
Jilin	26	43	81	125	151	255	0.2	0.2	0.4	0.4	0.3	0.1
Jiangsu	24	31	54	112	315	559	0.02	0	0	0	0	0
Jiangxi	0	0	0	0	0	0	0	0	0	0	0	0
Liaoning	0	0	0	0	0	0	0.1	0.06	0	0	0	0
Inner Mongolia	30	36	59	99	160	192	0.07	0.08	0.1	0.2	0.2	0.3
Ningxia	0	1	1	2	2	2	0.001	0.0002	0	0	0	0
Qinghai	32	45	122	58	120	158	0.1	0.04	0	0	0	0
Shandong	0.04	0.03	0.006	0	0	0	0.2	0.1	0.07	0.003	0	0
Shanxi	4	4	5	9	15	21	0	0	0	0	0	0
Shaanxi	2	2	3	6	5	6	0.1	0.1	0.2	0.5	0.4	0.5
Shanghai	0	0	0	0	0	0	0	0	0	0	0	0
Sichuan	48	62	79	97	137	242	0.06	0.08	0.10	0.1	0.2	0.3
Tianjin	2	2	2	3	4	5	0.01	0	0	0	0	0
Tibet	0	0	0	0	0	0	0	0	0	0	0	0
Xinjiang	99	120	143	226	461	558	0.2	0.2	0.2	0.4	0.7	0.8
Yunnan	0	0	1	1	2	6	0	0	0	0	0	0
Zhejiang	0	0	0	0	0	0	0	0	0	0	0	0
Chongqing	3	4	6	10	18	15	0.004	0.005	0.008	0.01	0.02	0.02

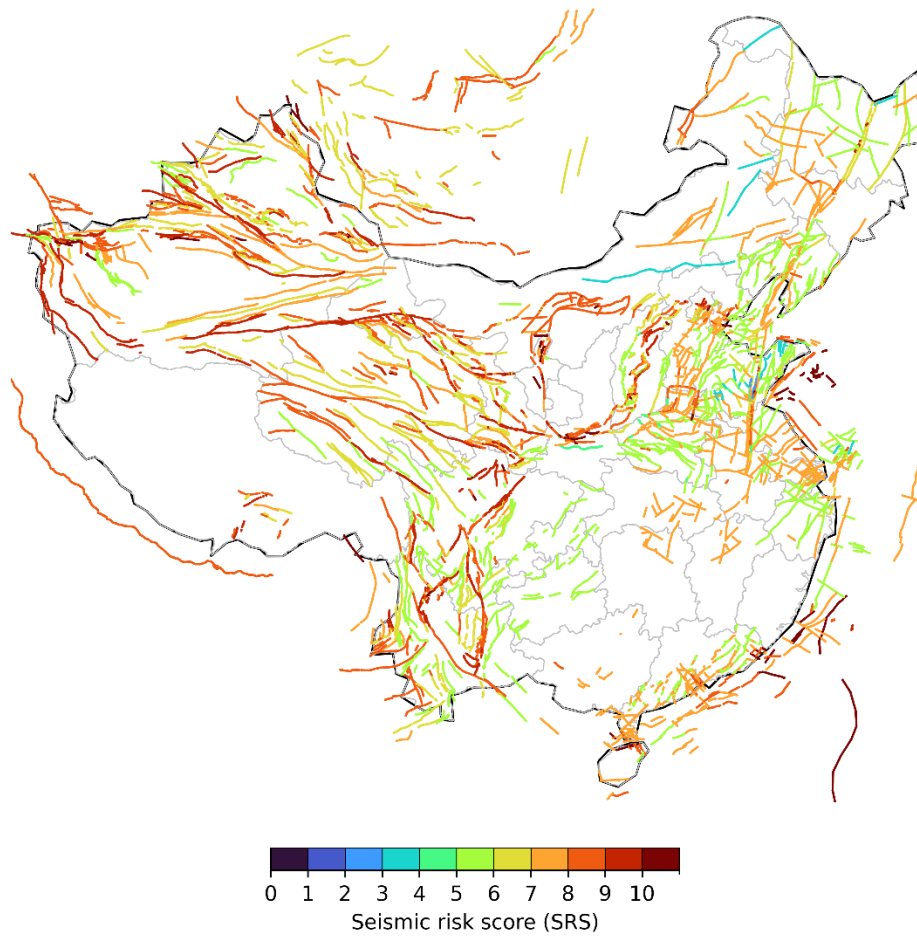
Supplementary Table 14: Comparison of CCS component unit cost with reference to China. 1 USD = 0.87 EUR.

Year	China (USD/ton)	USA (USD/ton)	EU (EUR/ton)
Capture Cost (USD/ton)	32.02 - 43.16	25 - 30 [31]	40-90 [32]
Transportation (USD/ton.km)	0.11 - 0.19	25 - 35 [31]	2-30 [32]
Storage cost (USD/ton)	6.96 - 8.35	25 - 35 [31]	5-35 [32]

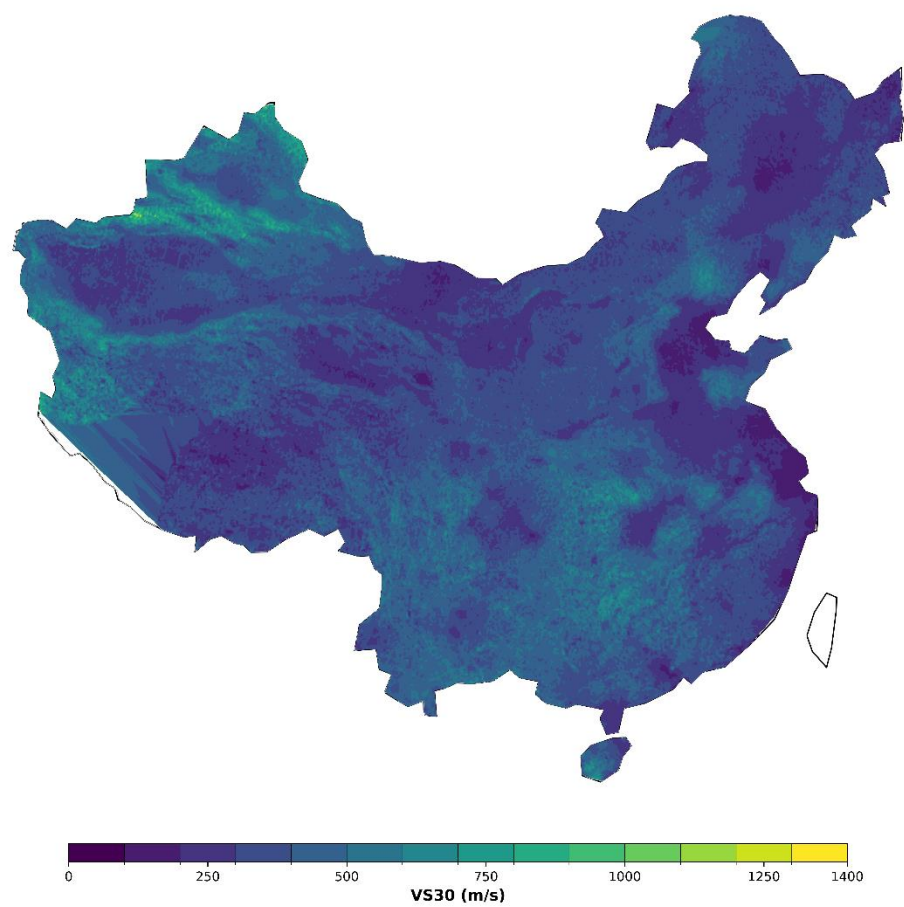
Supplementary Figures



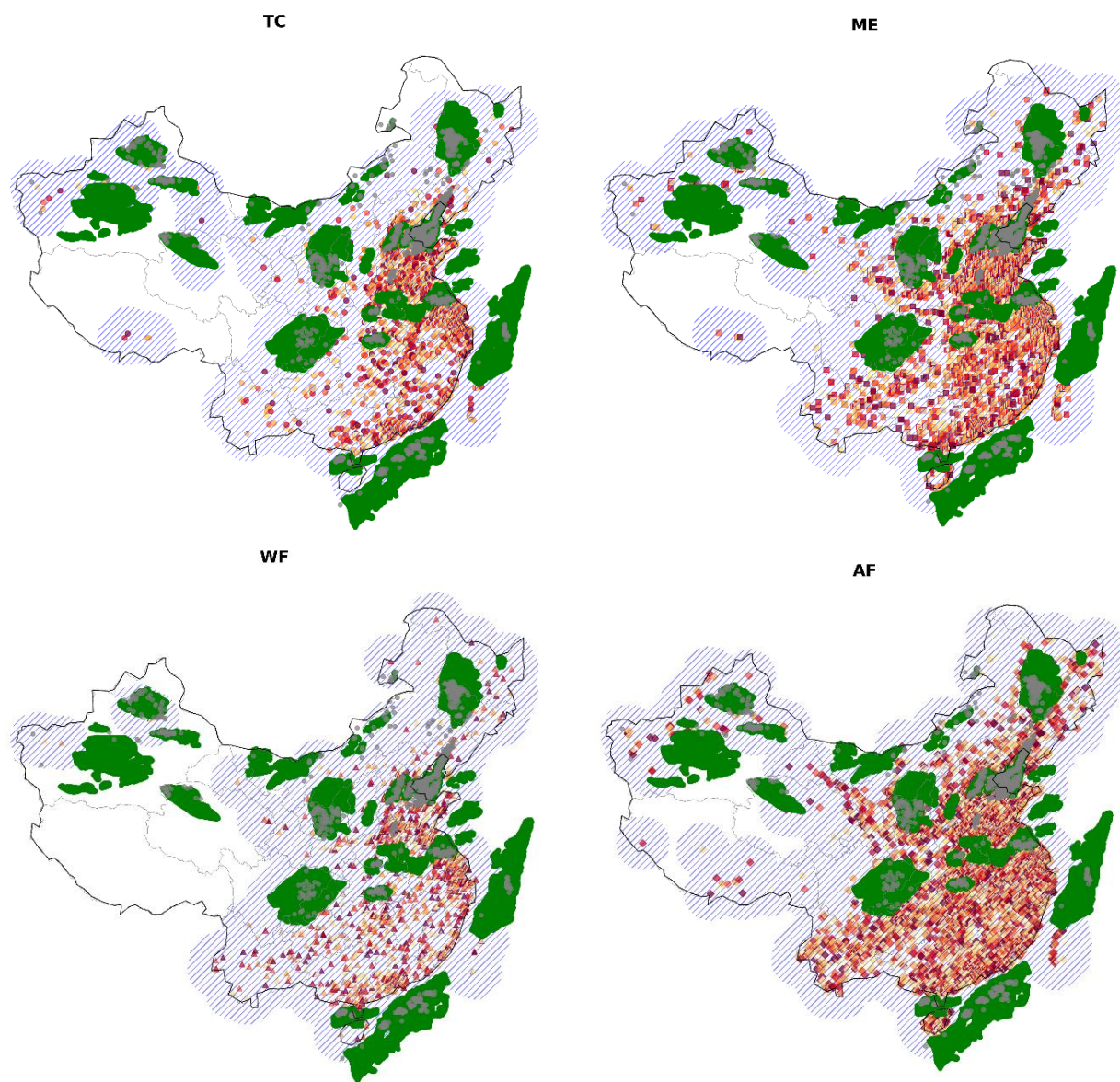
Supplementary Figure 1: Peak ground acceleration (PGA) at (a) 0.5%, (b) 10%, (c) 5%, (d) 2%, (e) 1%, and (f) 63% probability of exceedance in 50 years.

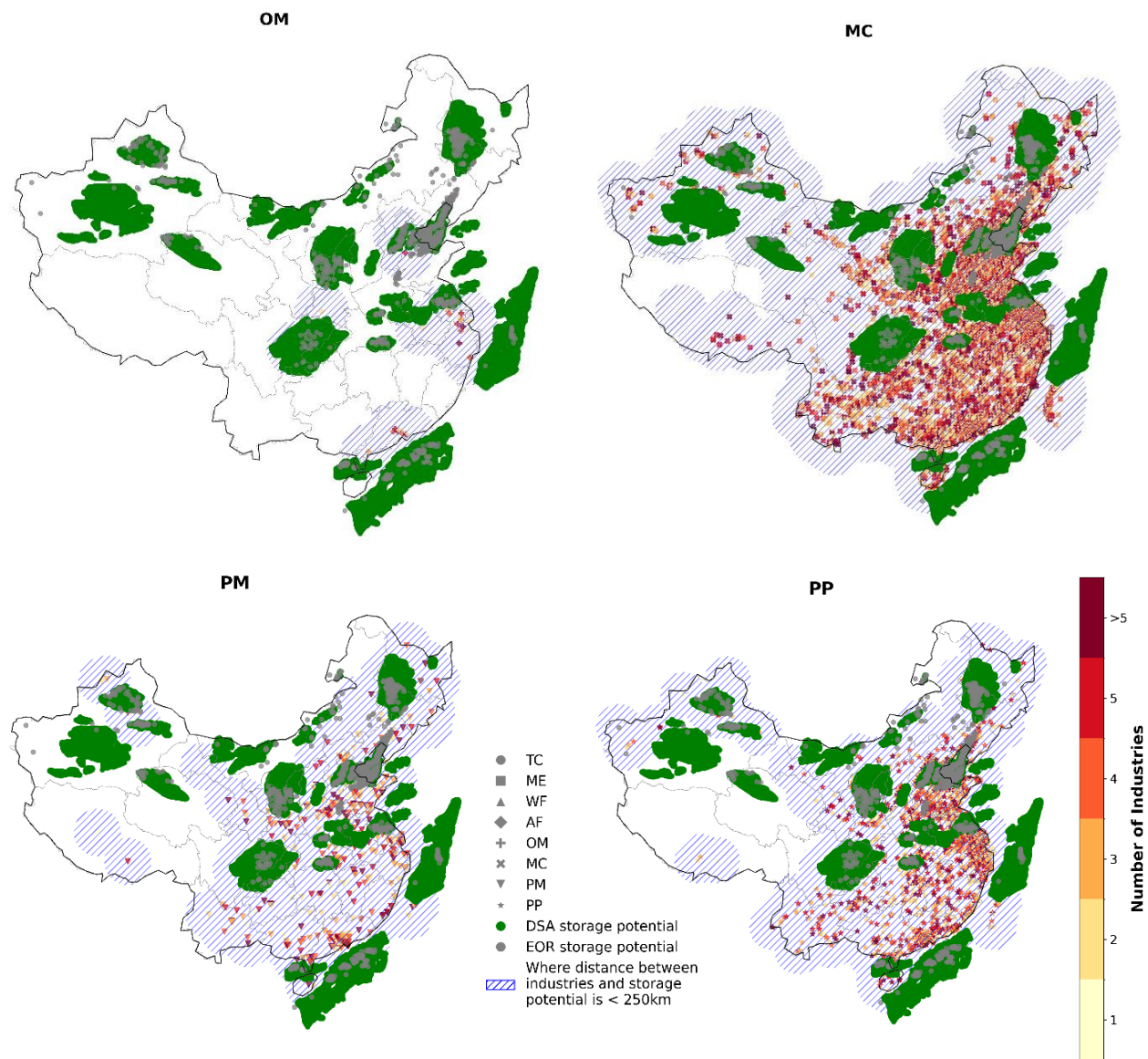


Supplementary Figure 2: Fault activity risk score of active fault lines.

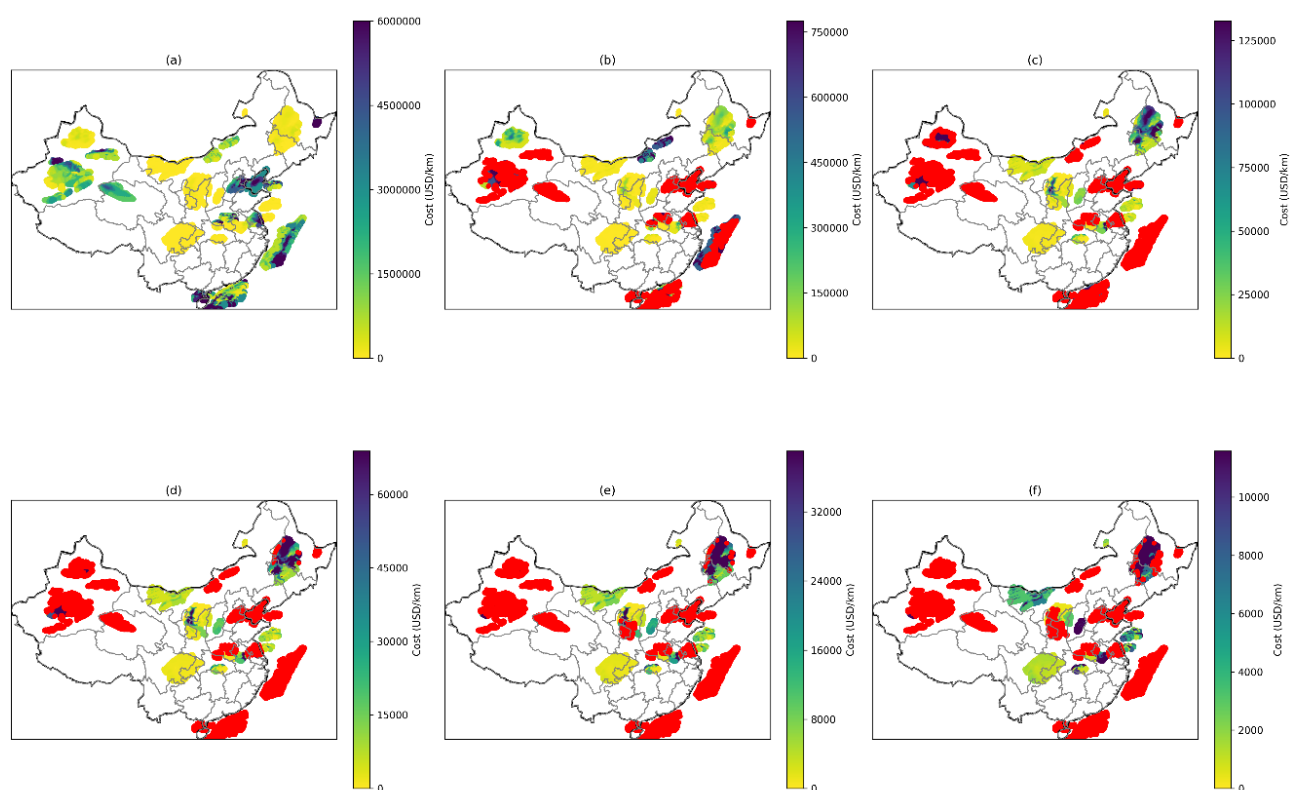


Supplementary Figure 3: Vs30 map for China at 5km resolution [37].

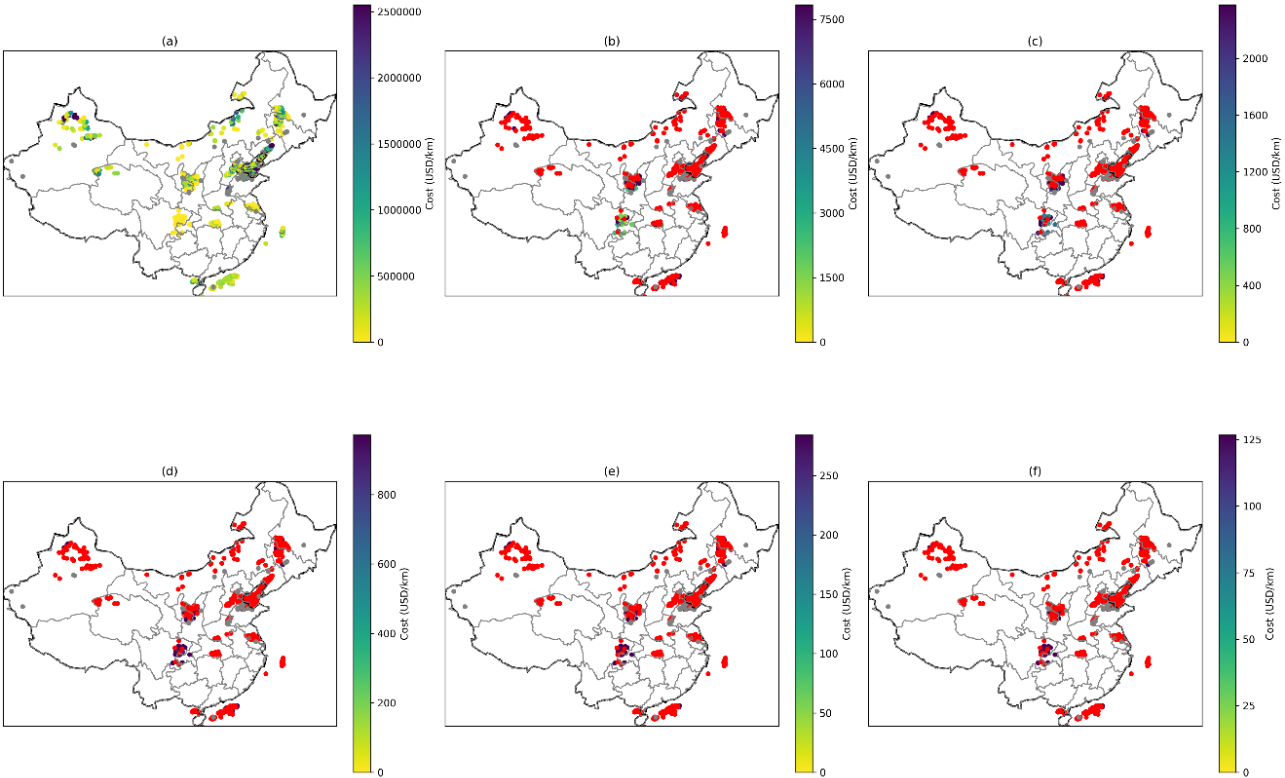




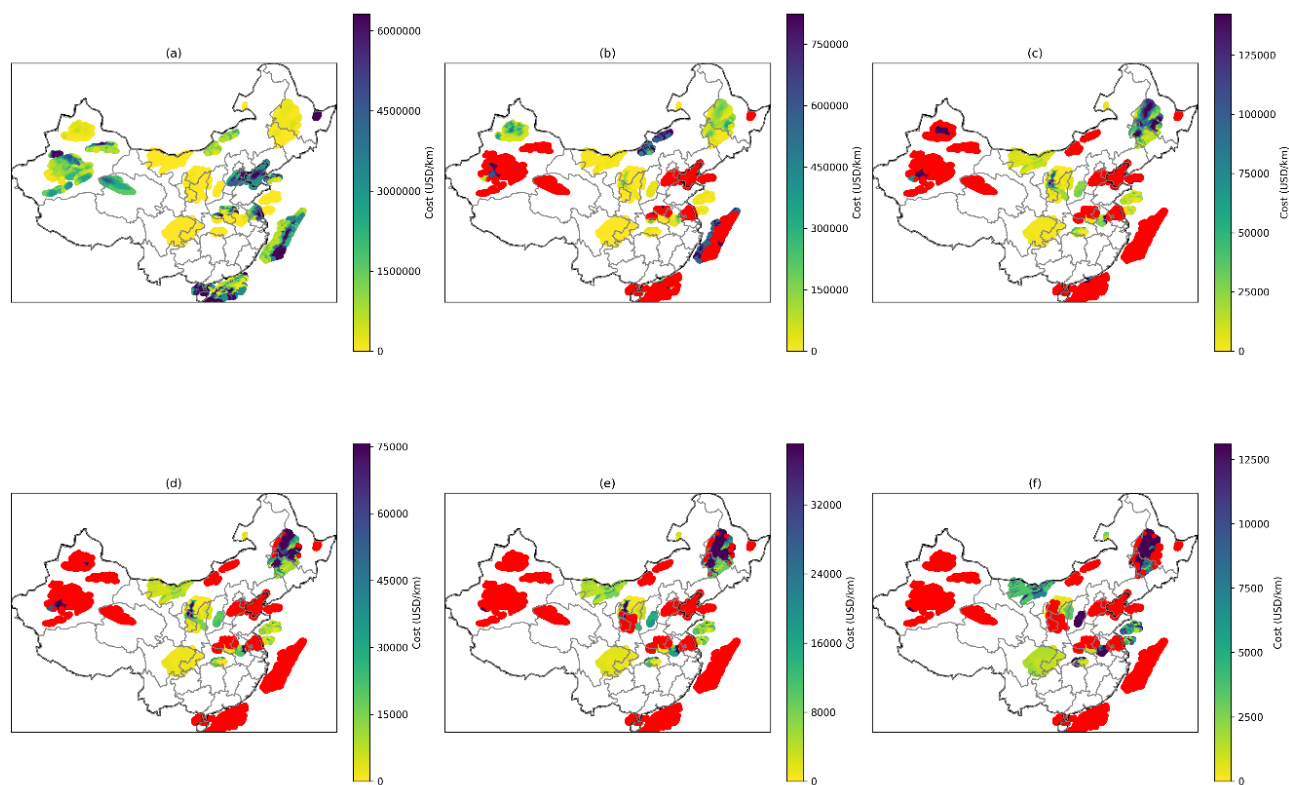
Supplementary Figure 4: The distribution of CO₂ sources (industrial clusters) and storage in China. The purple-hatched shaded regions denote industrial facilities situated within a 250 km radius of potential storage locations, representing optimal candidates for CCS implementation using pipeline transportation infrastructure. TC: Textile and garment; ME: mechatronics and equipment; WF: wood furniture; AF: Food processing firms of agricultural and sideline products; MC: metallurgical, chemical, and resource rough processing; PM: pharmaceutical; PP: paper making; and OM: other manufacturing firms represent the classification of manufacturing industries



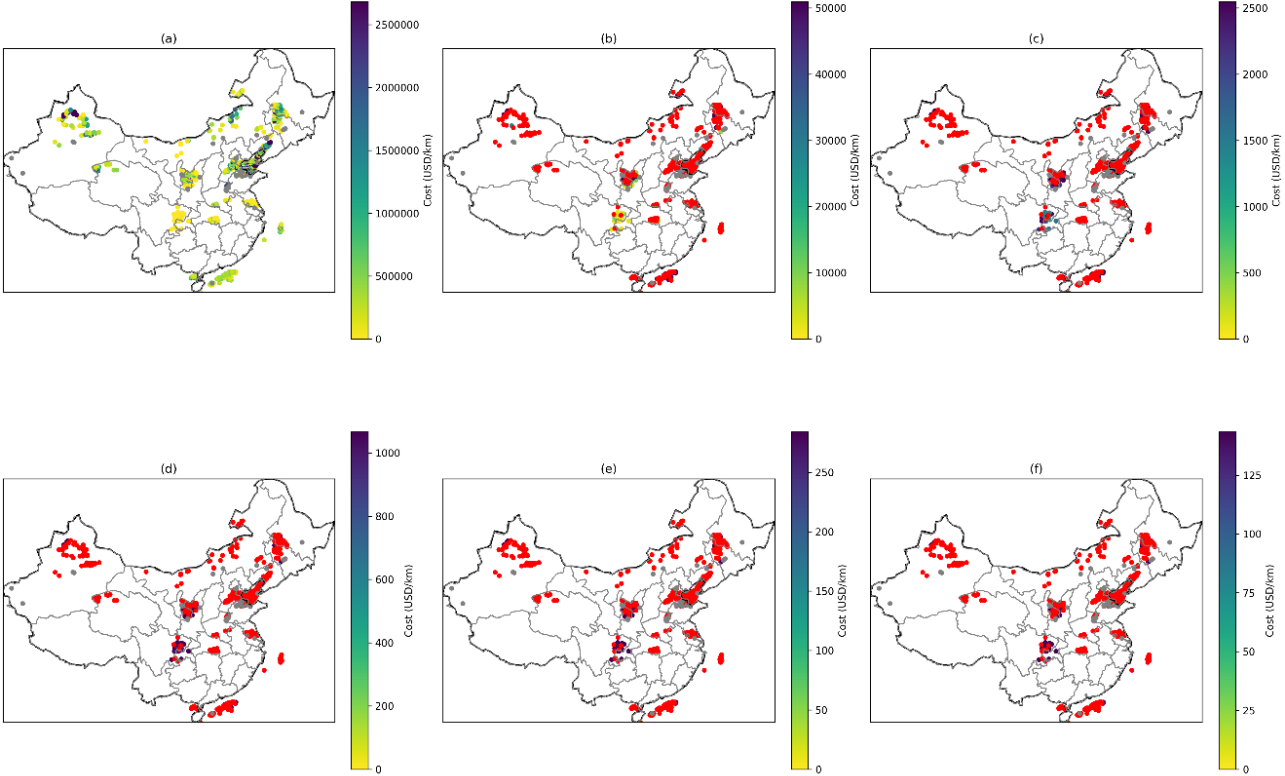
Supplementary Figure 5: A 5 km x 5 km spatial distribution of the DSA minimum carbon removal cost using the pipeline mode of transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The red-shaded areas indicate areas where the storage potential has been fully utilized/exhausted.



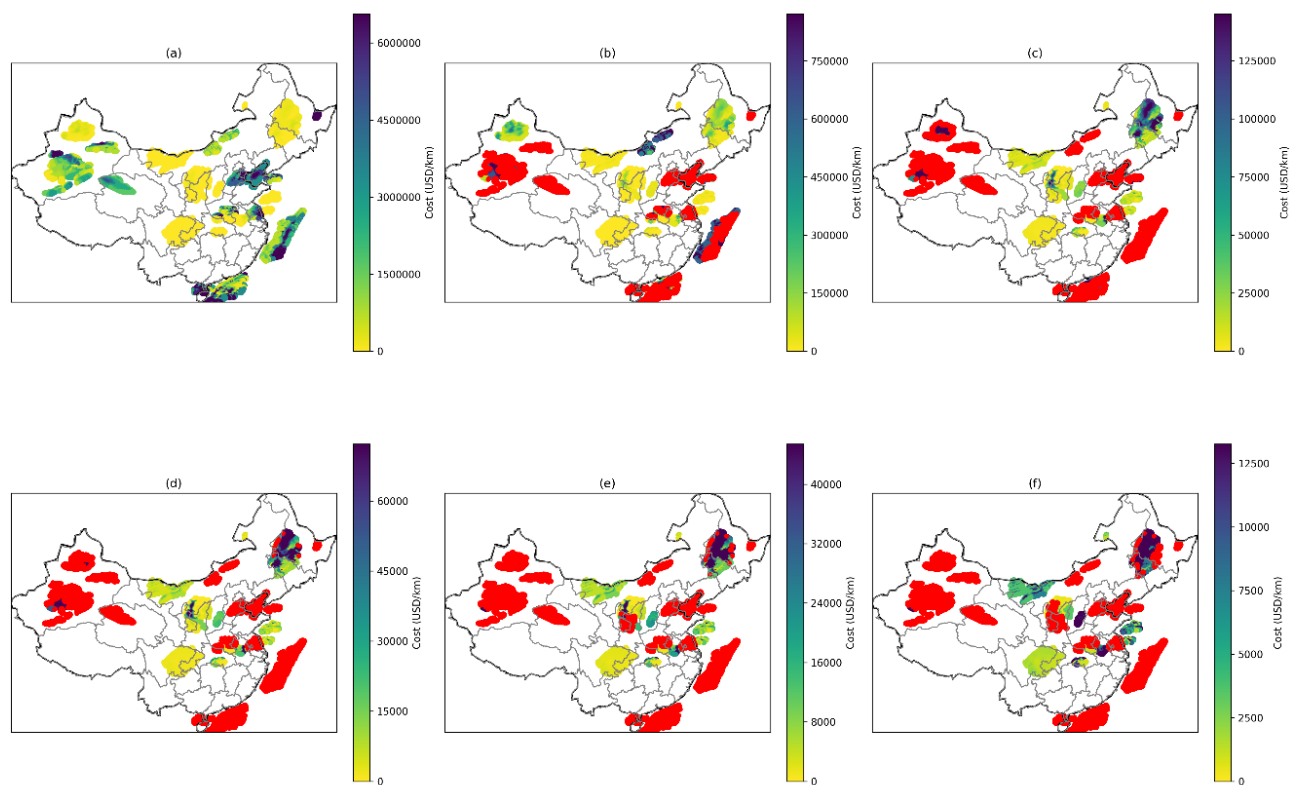
Supplementary Figure 6: A 5 km x 5 km spatial distribution of the EOR minimum carbon removal cost using the pipeline mode of transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The gray-shaded region is as explained in Supplementary Figure 17, while the red-shaded areas are as explained in Supplementary Figure 5.



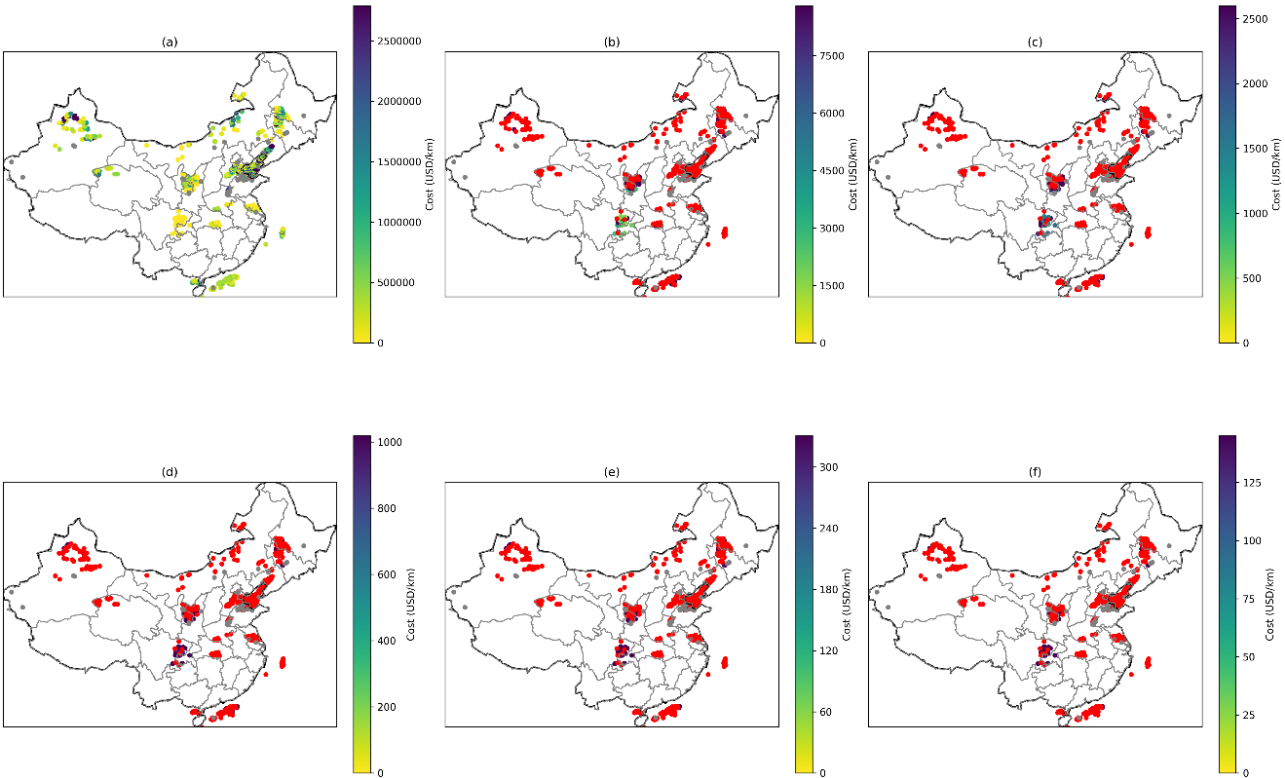
Supplementary Figure 7: A 5km x 5 km spatial distribution of the DSA minimum carbon removal cost using the tank mode of transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The red-shaded areas indicate areas where the storage potential has been exhausted.



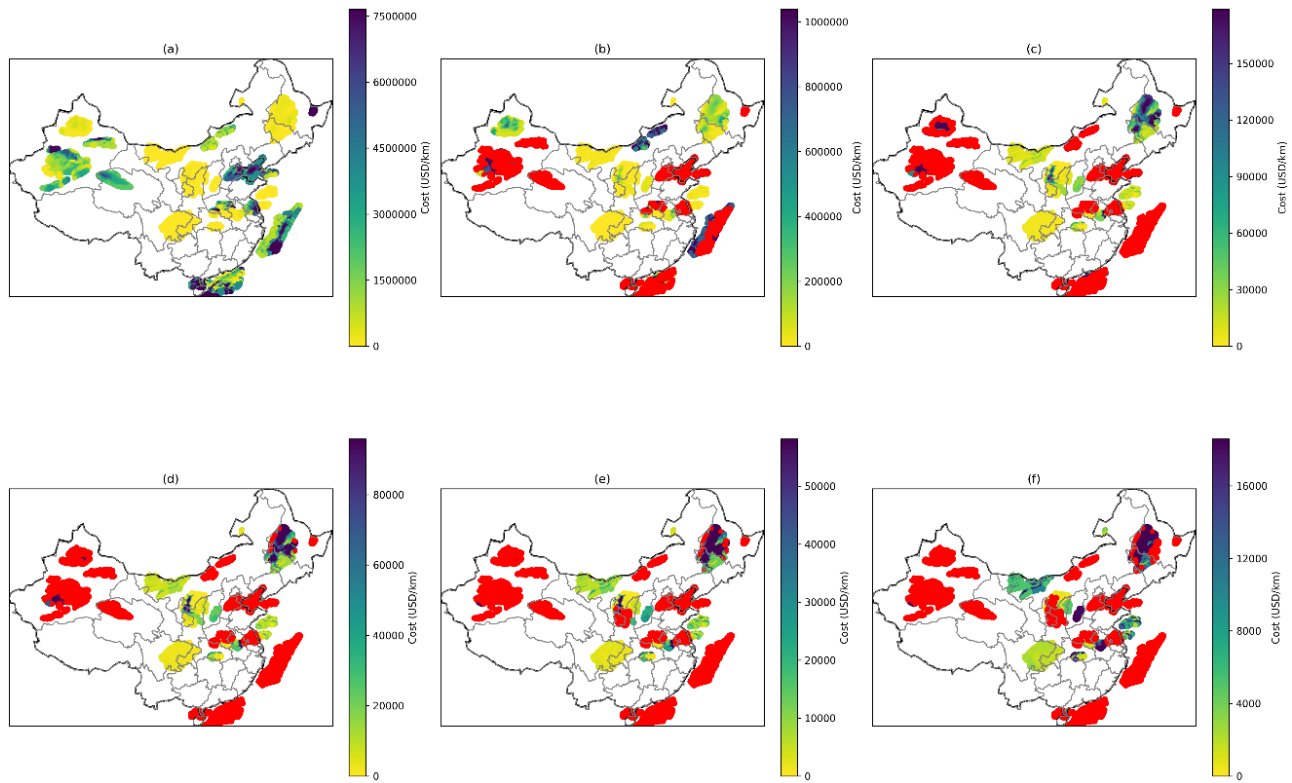
Supplementary Figure 8: A 5km x 5 km spatial distribution of the EOR minimum carbon removal cost using the tank mode of transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The gray-shaded region is as explained in Supplementary Figure 17, while the red-shaded areas are as explained in Supplementary Figure 5.



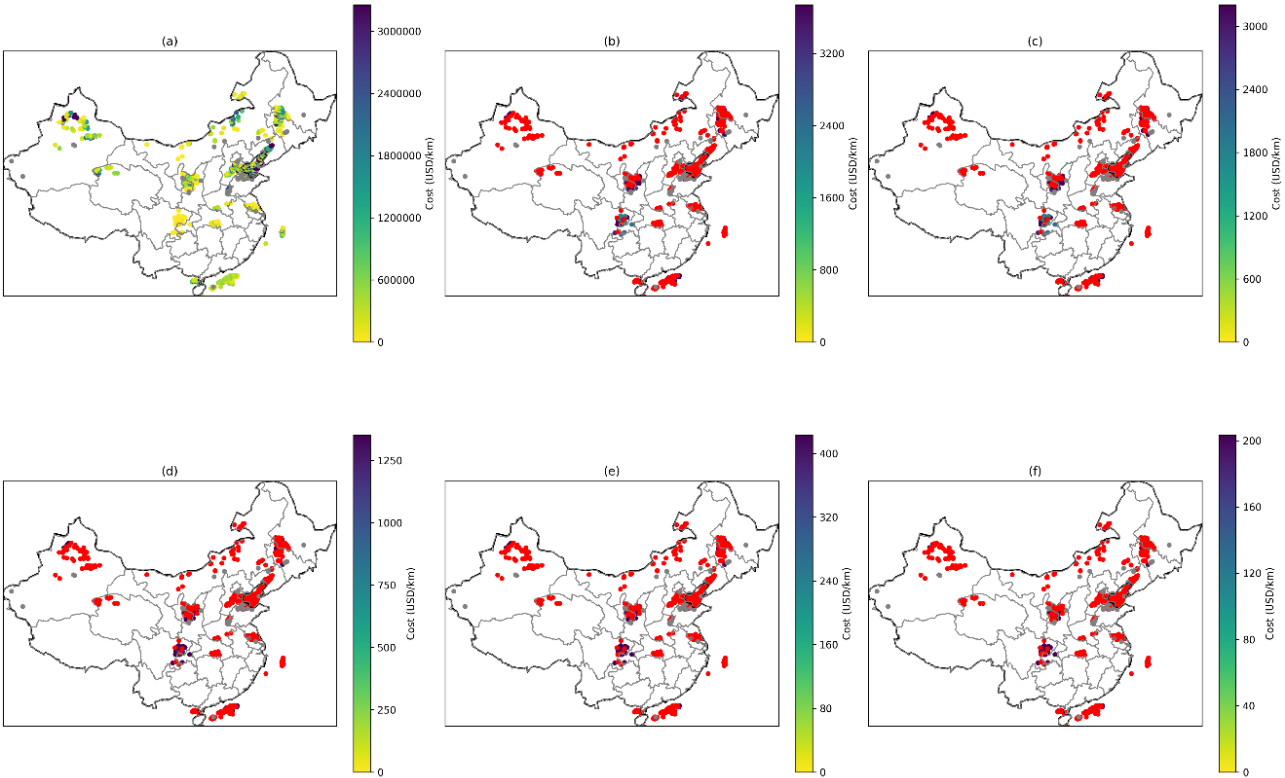
Supplementary Figure 9: The average carbon removal cost for DSA using the pipeline mode of transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The red-shaded areas indicate areas where the storage potential has been exhausted.



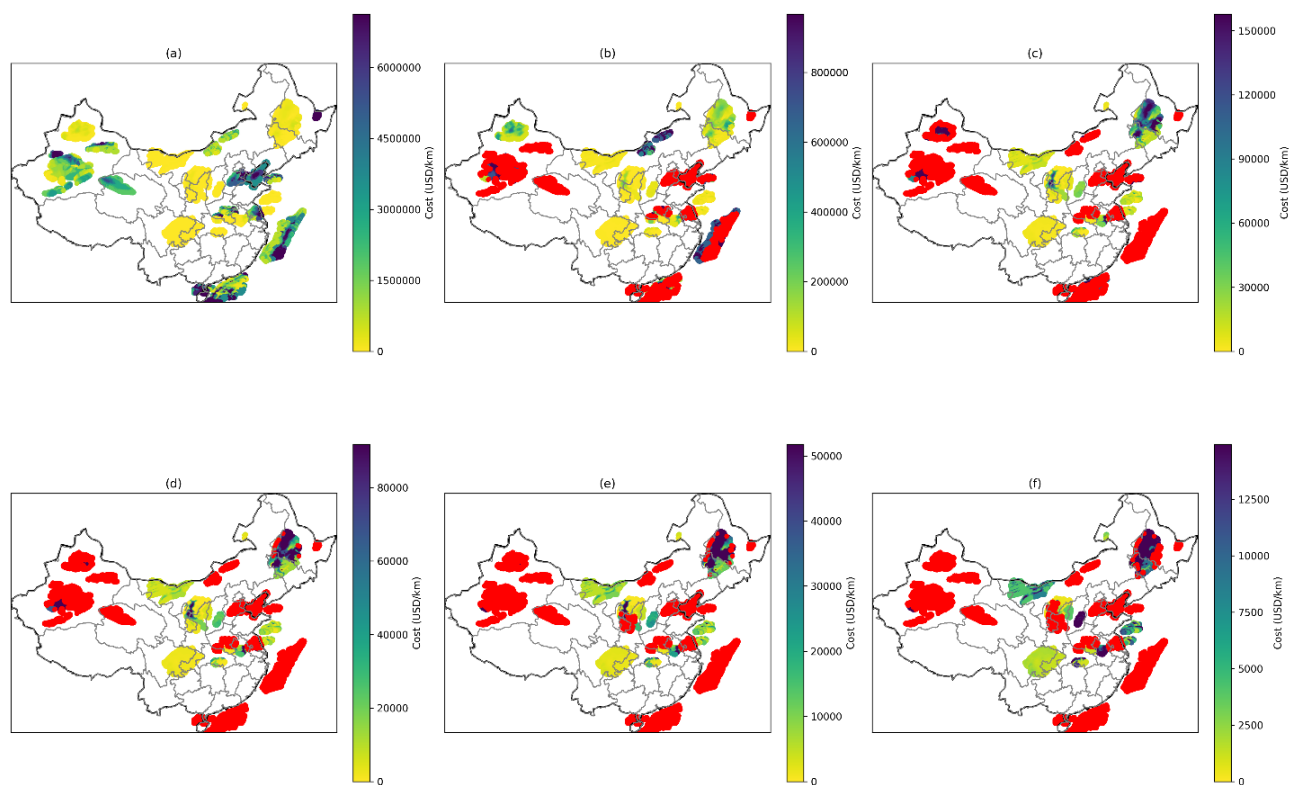
Supplementary Figure 10: The average carbon removal cost for EOR using the pipeline mode of transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The gray-shaded region is as explained in Supplementary Figure 17, while the red-shaded areas are as explained in Supplementary Figure 5.



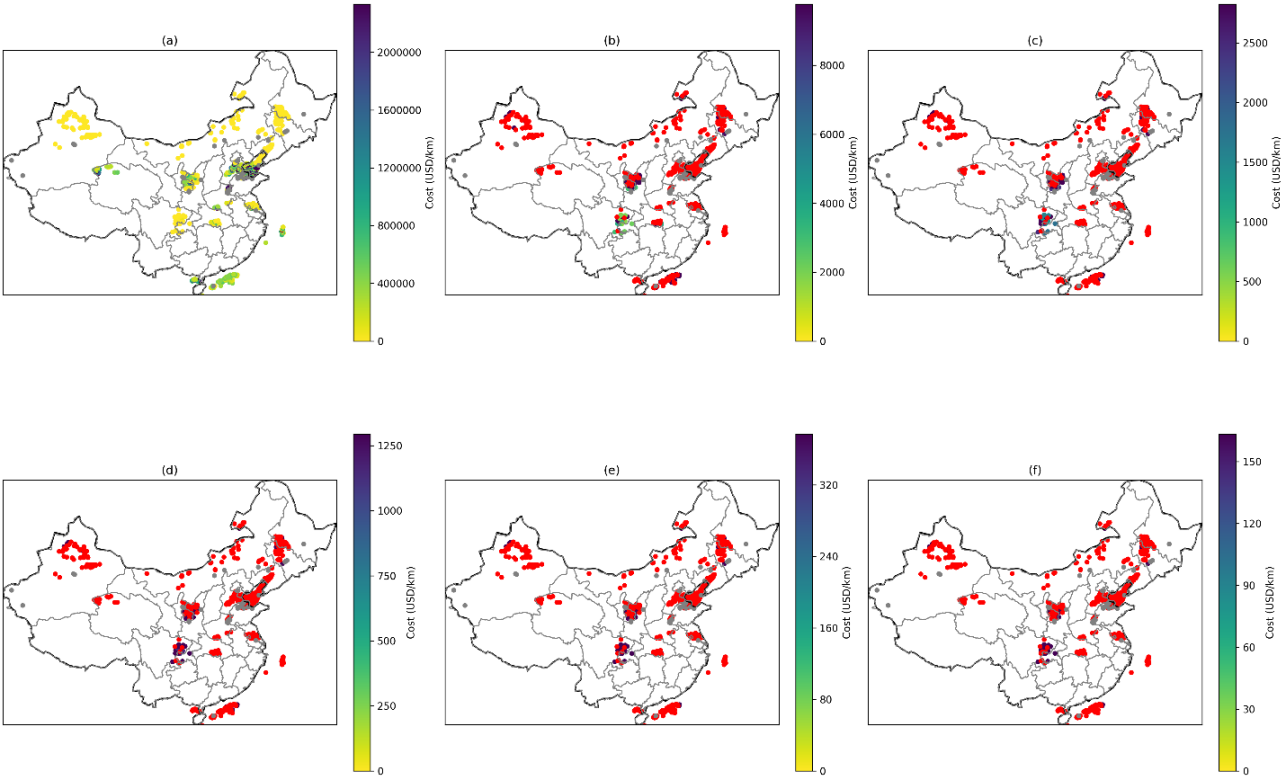
Supplementary Figure 11: The average carbon removal cost for DSA using the tank mode of transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The red-shaded areas indicate areas where the storage potential has been exhausted.



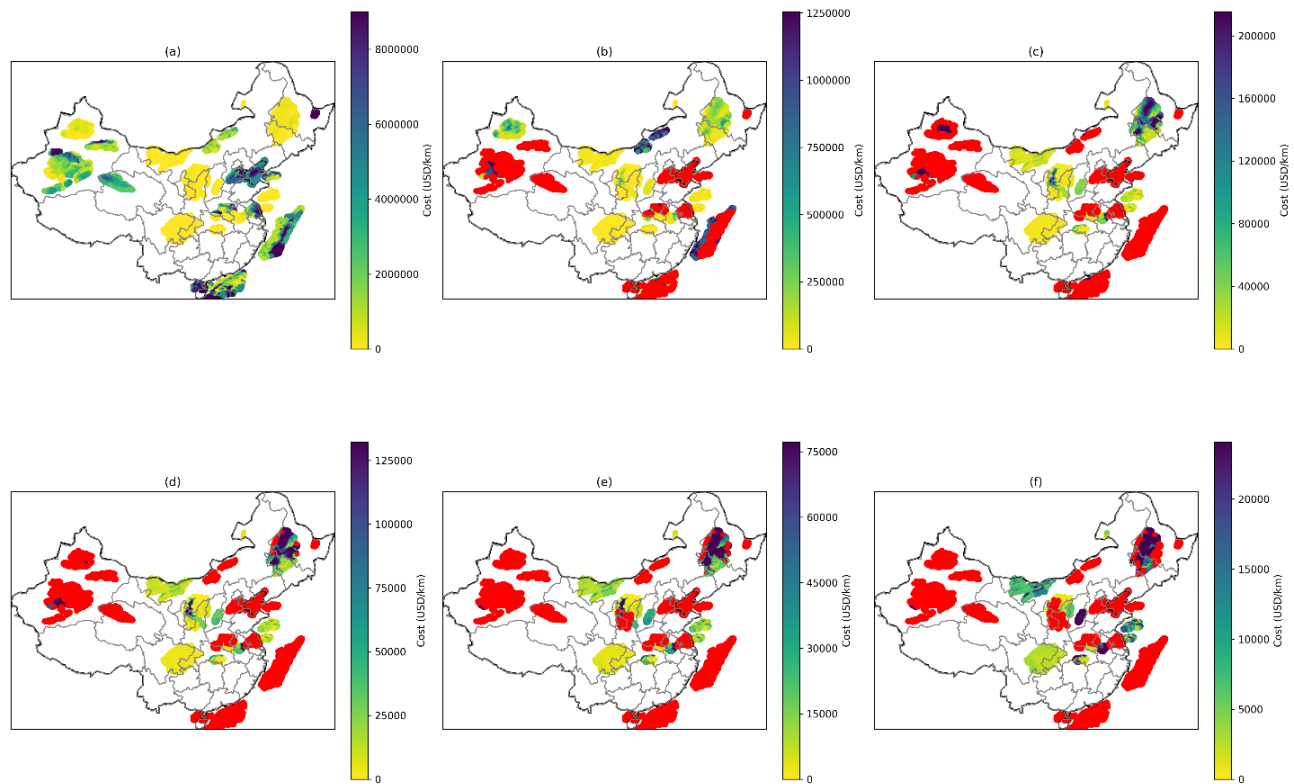
Supplementary Figure 12: The average carbon removal cost for EOR using the tank mode of transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The gray-shaded region is as explained in Supplementary Figure 17, while the red-shaded areas are as explained in Supplementary Figure 5.



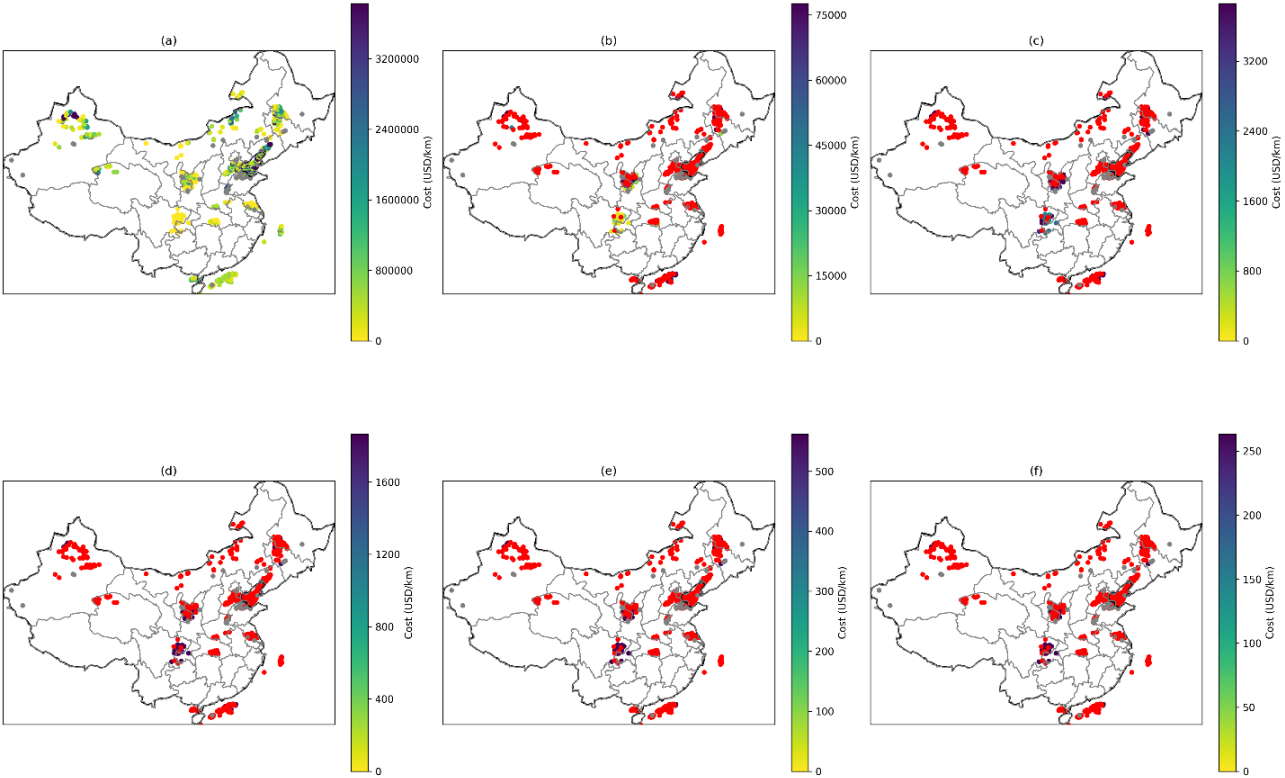
Supplementary Figure 13: The maximum carbon removal cost for DSA using the pipeline mode of transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The red-shaded areas indicate areas where the storage potential has been exhausted.



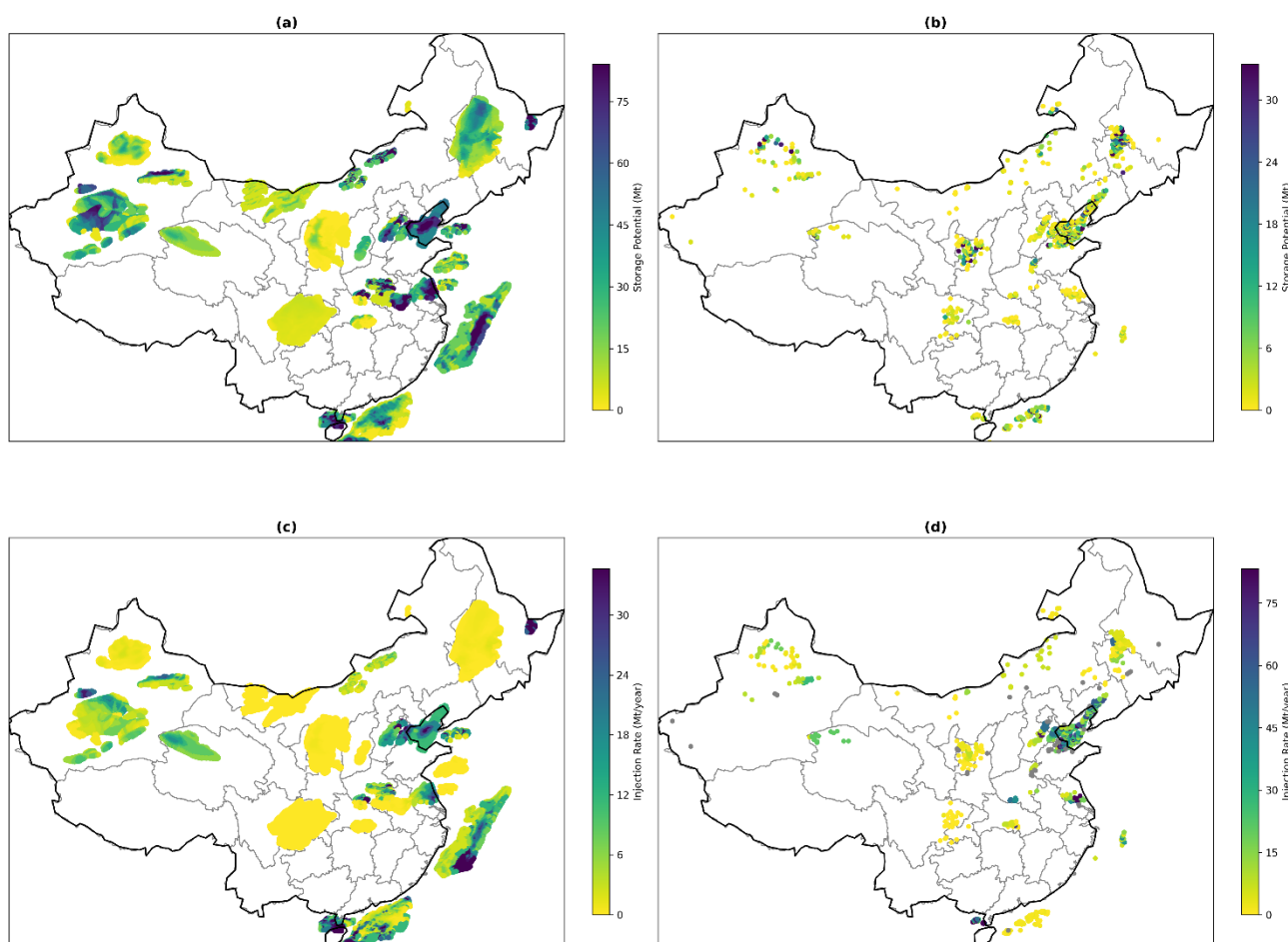
Supplementary Figure 14: The maximum carbon removal cost for EOR using the pipeline mode of transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The gray-shaded region is as explained in Supplementary Figure 17, while the red-shaded areas are as explained in Supplementary Figure 5.



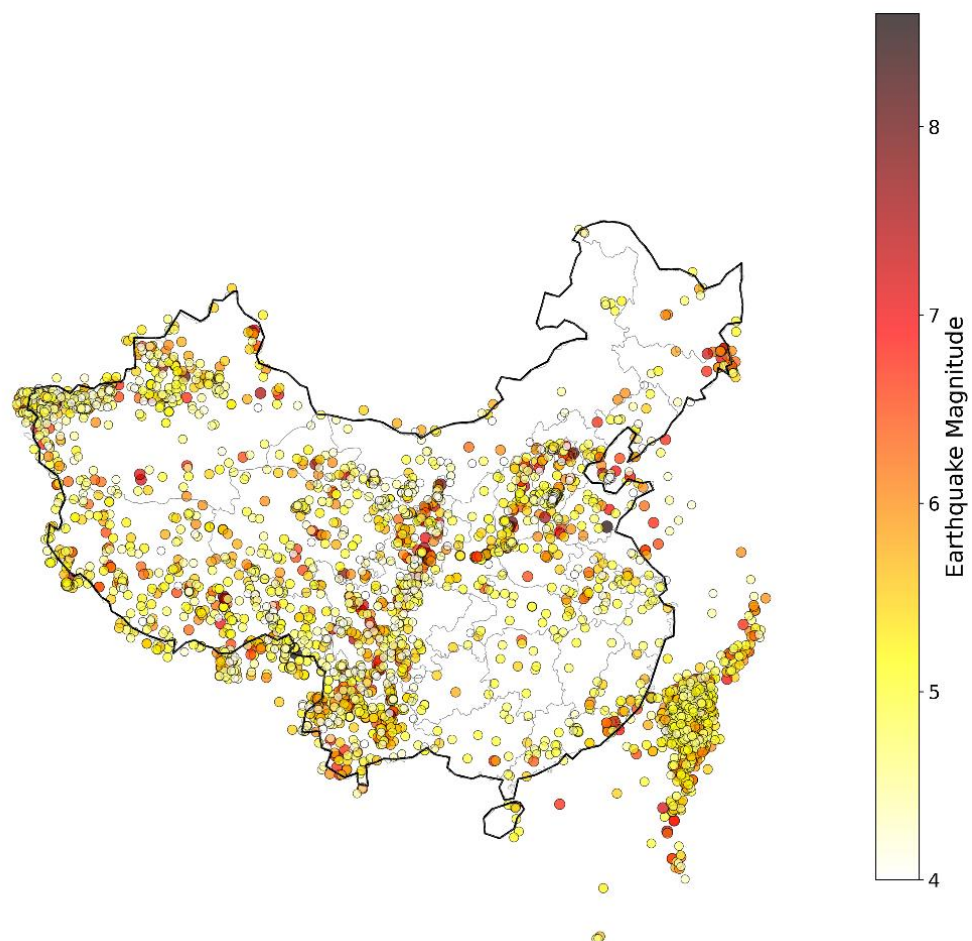
Supplementary Figure 15: The maximum carbon removal cost for DSA using the tank transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The red-shaded areas indicate areas where the storage potential has been exhausted.



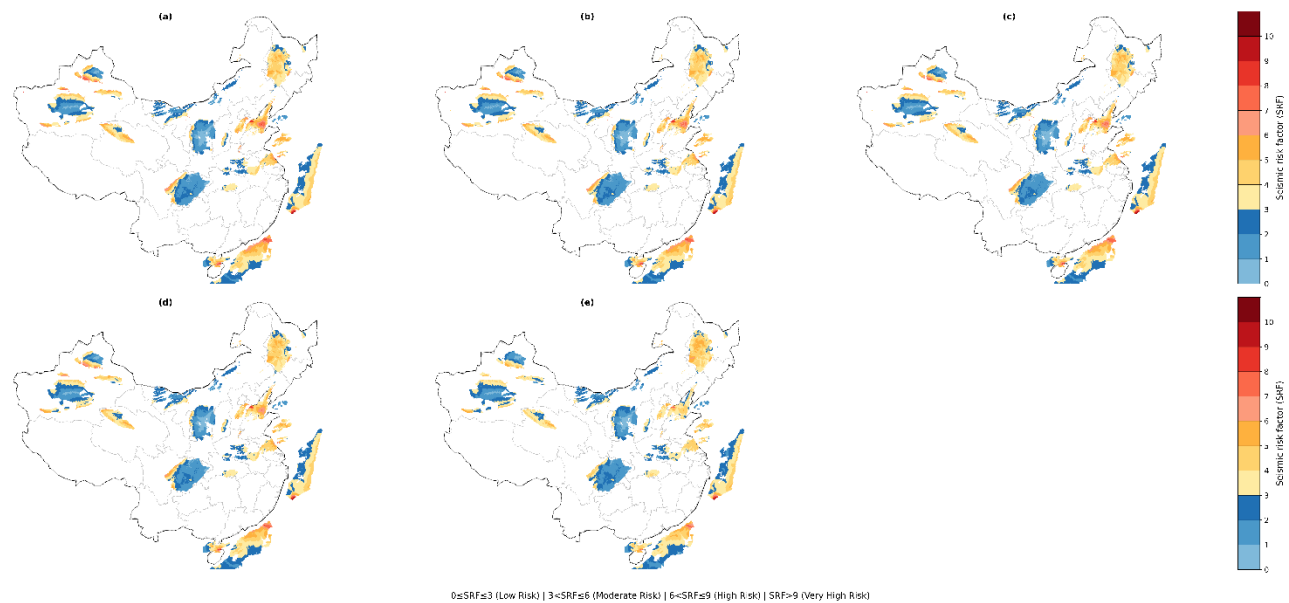
Supplementary Figure 16: The maximum carbon removal cost for EOR using the tank transport for (a) 2025, (b) 2030, (c) 2035, (d) 2040, (e) 2045, and (f) 2050. The gray-shaded region is as explained in Supplementary Figure 17, while the red-shaded areas are as explained in Supplementary Figure 5.



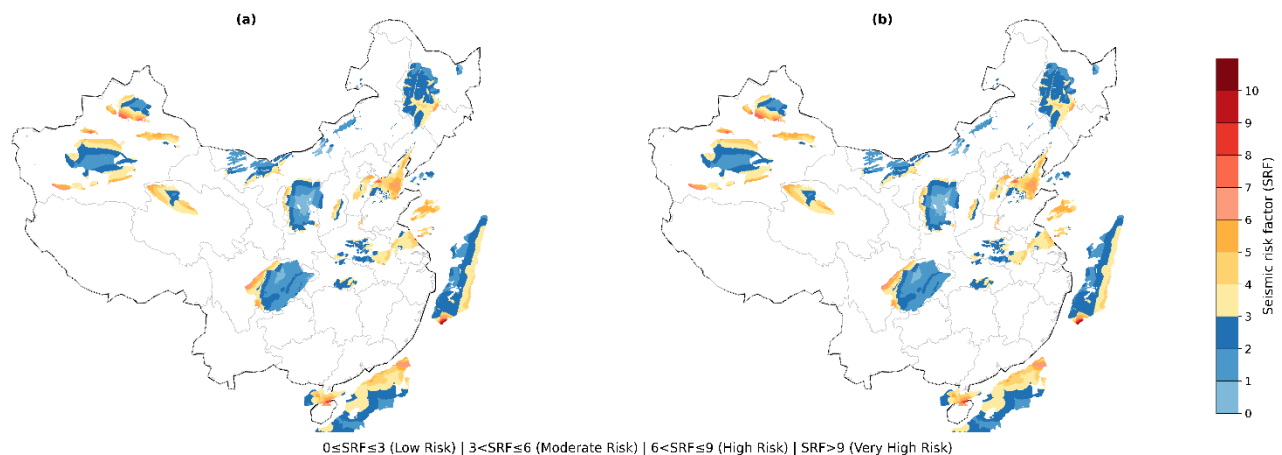
Supplementary Figure 17: A 5 km x 5 km spatial distribution of the storage potential for (a) deep saline aquifers (DSA) and (b) enhanced oil recovery (EOR). (c) and (d) show the spatial distribution of the injection rate capacity for DSA and EOR, respectively. The gray-shaded regions represent areas where storage potential remains untapped due to undetermined injection rates at those specific grid points.



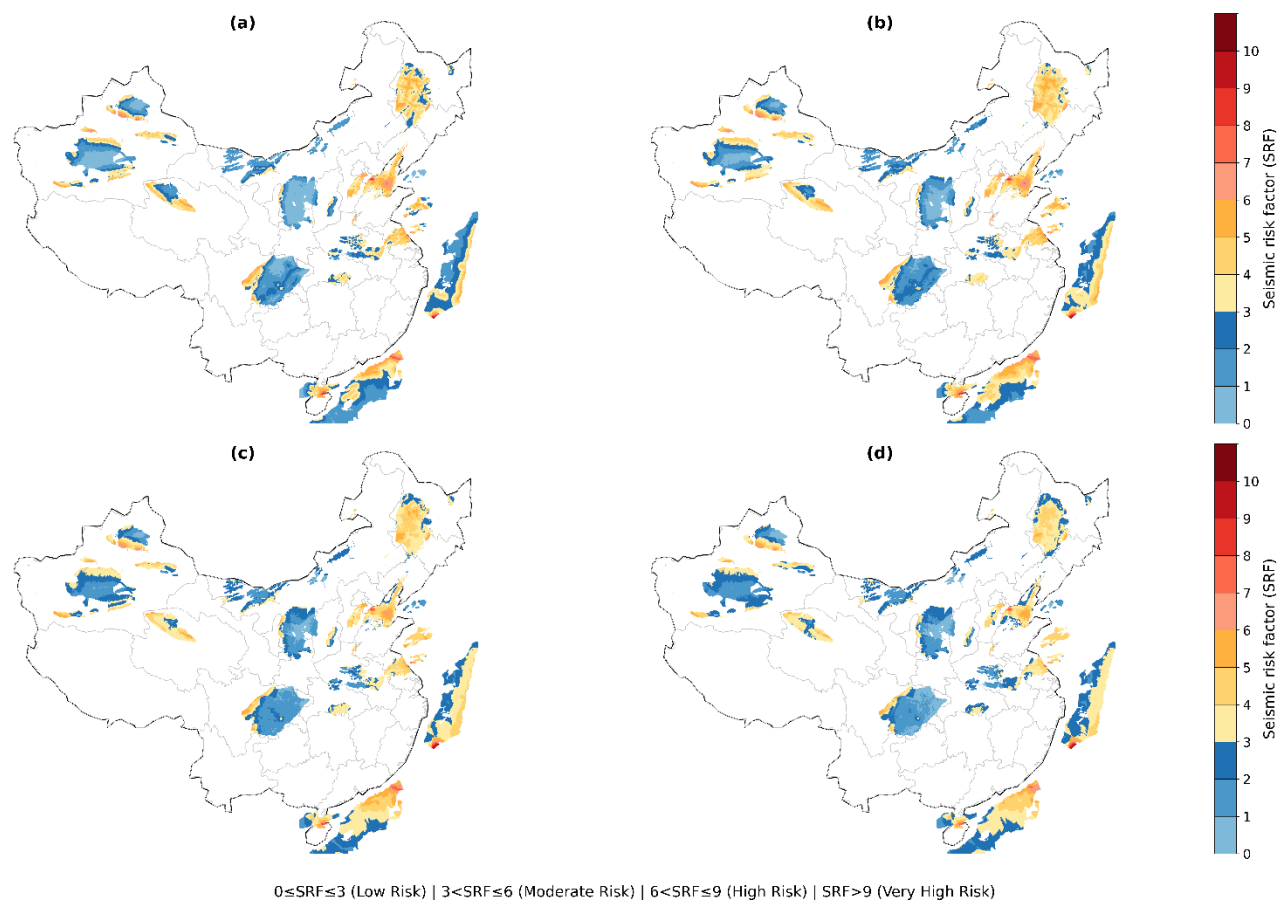
Supplementary Figure 18: Historic earthquake events distribution based on magnitude.



Supplementary Figure 19: Seismic risk factor at (a) 0.5%, (b) 1%, (c) 2%, (d) 5%, and (e) 63% PGA probability levels of exceedance in 50 years. PGA exceedance probabilities are systematically varied across five available levels while holding fault proximity ($\lambda = 200$ km) and site amplification constant.

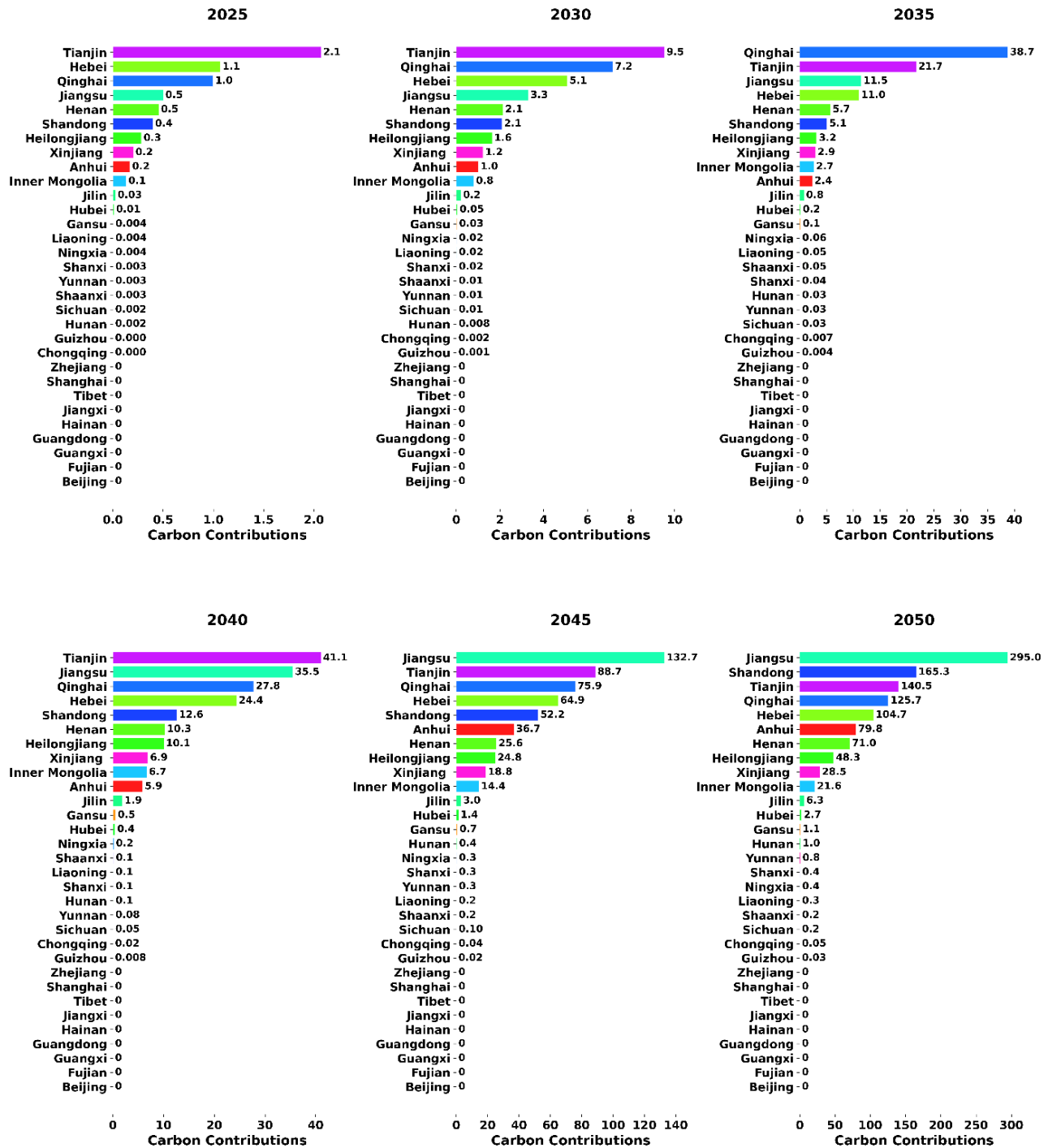


Supplementary Figure 20: The seismic risk factor using (a) GB 50011-2010 code and (b) no amplification while maintaining a constant PGA probability level (10%) and fault proximity ($\lambda = 200$ km).

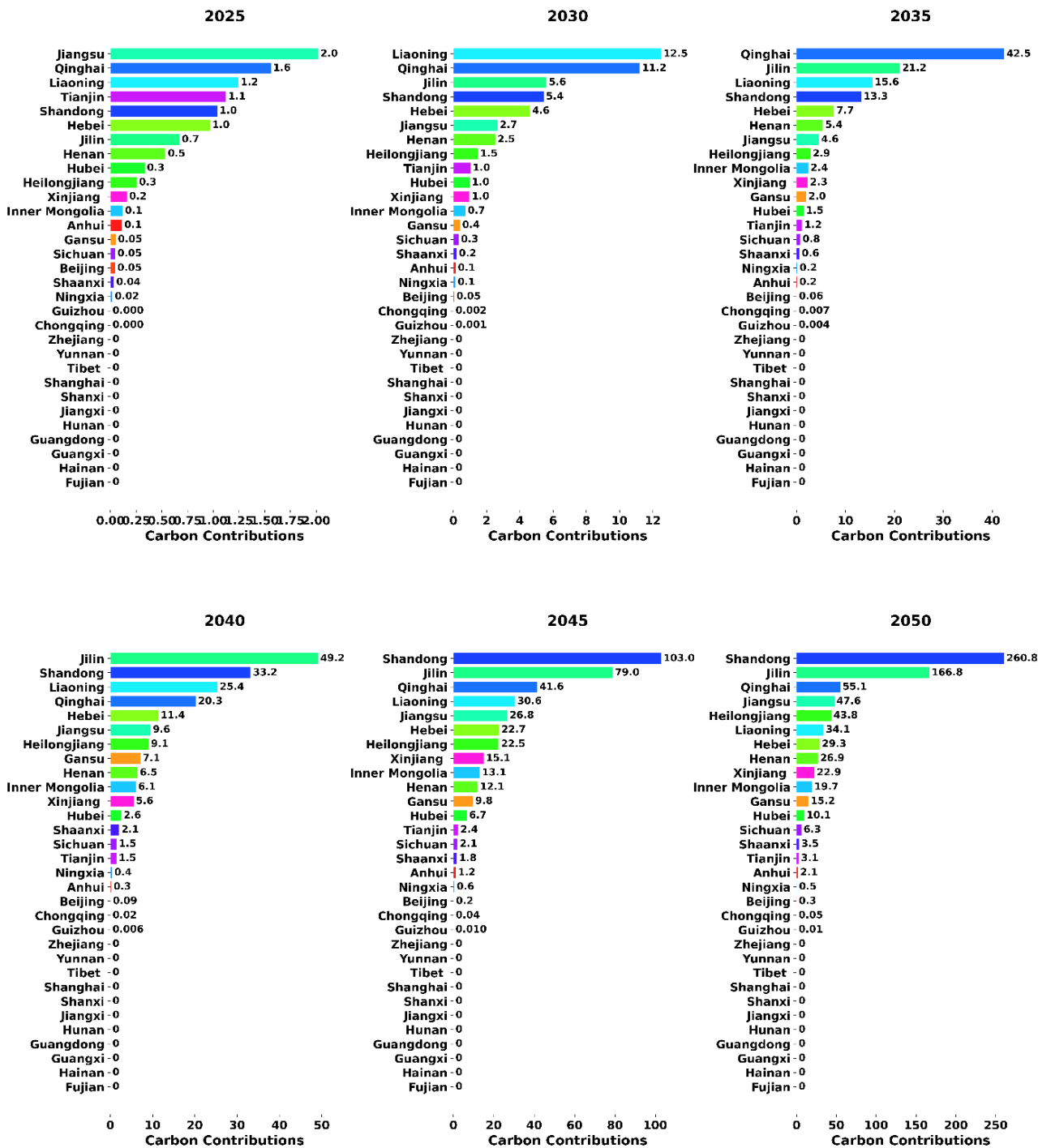


Supplementary Figure 21: The seismic risk factor at varying decay constant at (a) $\lambda = 50$ km, (b) $\lambda = 100$ km, (c) $\lambda = 300$ km, and (d) $\lambda = 500$ km while maintaining a constant PGA probability level (10% in 50 years exceedance) and site amplification.

(a)



(b)



Supplementary Figure 22: The percentage contribution of CCS to emission reduction in each province for (a) DSA and (b) EOR.

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